# Physiological Linkage and Interpersonal Outcomes: A Systematic Review and Meta-Analysis of Methodological and Contextual Differences

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**Abstract**

Strong, supportive interpersonal relationships play an important role in health and well-being. Physiological linkage, or the relationship between interacting individuals' physiological responses, offers a potential way to capture interpersonal dynamics continuously and passively, complementing subjective and behavioral assessments. However, studies report mixed associations between linkage and interpersonal outcomes. Different physiological measures and analytical approaches capture different aspects of linkage, and it remains unclear how these choices, alongside differences in interaction contexts and outcomes, relate to the mixed findings. We conducted a systematic review and meta-analysis to examine these associations across physiological measures, linkage definitions and methods, interaction contexts, and outcome categories. We found small, nonsignificant average associations for both unadjusted and adjusted effects, alongside substantial variation across effects. A restricted analysis of unadjusted effects showed a more positive association for sympathetic than parasympathetic measures, while most moderator analyses provided limited evidence to explain the variation. These findings do not support treating greater physiological linkage as a general indicator of better interpersonal outcomes. Instead, they highlight the need for clear hypotheses connecting physiological measures and analytical choices to specific interpersonal processes, and for systematic method comparisons within the same datasets. This work contributes a quantitative synthesis of the evidence, a structured account of methodological differences, and an open-source literature database to support replication, alternative analyses, and future updates.

## 1 Introduction

Forming and maintaining strong, positive, and stable interpersonal relationships is a fundamental human need [4]. According to the social baseline theory, engaging in such relationships is advantageous for human survival because it decreases the risk of threats and distributes effortful executive functions such as self-regulation and decision-making [5]. Having a supportive partner or family contributes to psychological well-being [70], and building warm, intimate, and stable relationships with one's mother is critical for children to have good mental health [25]. On the other hand, the lack of human connection leads to undesired outcomes. For example, the lack of supportive social relationships is associated with higher mood disturbances on and after stressful days [19]. Not limited to stress, unsatisfactory marital relationships have been associated with various adverse health outcomes, including lower sleep quality and quantity, depression, obesity, cardiovascular diseases, diabetes, and cancer [38].

Understanding the characteristics of healthy relationship dynamics can be useful in designing relationship interventions. Relationship dynamics can be studied using self-reported measures, behavioral observations, and qualitative analyses; however, subjectivity and variability often limit these approaches. Moreover, collecting such data continuously is challenging, often resulting in discrete snapshots that may fail to capture how relationships evolve over time. Physiological measures, such as autonomic nervous system (ANS) activities [40], offer objective, quantifiable, and continuous metrics that may potentially capture complex

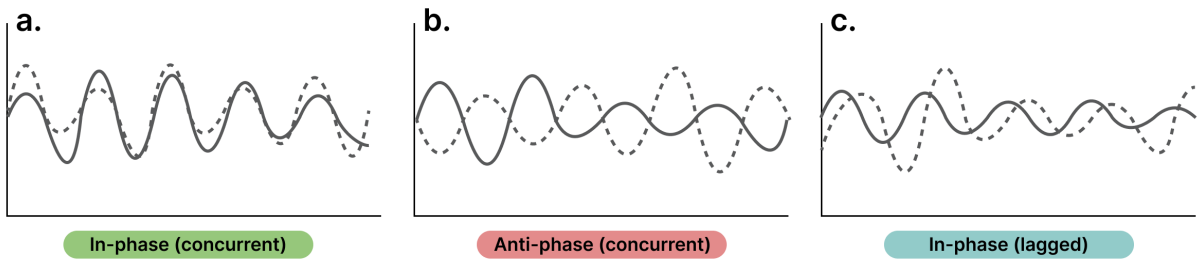



Figure 1: Different types of physiological linkage: (a) concurrent in-phase linkage, in which two signals change in the same direction at the same time; (b) concurrent anti-phase linkage, in which two signals change in opposite directions at the same time; and (c) lagged in-phase linkage, in which the two signals change in the same direction with a temporal offset. The two lines represent illustrative physiological signals from two individuals.

aspects of interaction dynamics that are not apparent only through subjective or behavioral assessments. If the physiological characteristics of healthy dynamics can be identified, it may also enable automatic and passive detection of the quality of relationships and tailored interventions in the future.

Physiological linkage—also called physiological synchrony, concordance, or coupling—is the relatedness between physiological responses of members in an interacting dyad [41], and it is a potential biomarker for relationship dynamics. As shown in Figure 1, two persons' physiology could change in the same (in-phase) or the opposite (anti-phase) directions, either concurrently or with temporal lags. A large body of research shows that physiological linkage occurs in various dyadic contexts, including romantic partners [41, 7, 16, 13, 72, 57], parent-child dyads [22, 24, 74], therapist-client relationships [35, 45], teammates [6, 52, 61], and even strangers [7, 60]. Some researchers have found that the level of strength of physiological linkage is associated with various relationship dynamics. For example, Marci et al. [45] examined therapists' and clients' linkage in skin conductance during psychotherapy sessions. They found that the linkage positively correlates with the perceived empathy of therapists rated by clients. Behrens et al. [6] measured skin conductance and heart rate while participants played the Prisoner's Dilemma game and found that the linkage in skin conductance was positively associated with the cooperative success of dyads. Woltering et al. [74] measured heart rate while children and mothers engaged in positive and negative discussions. They found that dyads with stronger physiological linkage showed more dyadic attunement (i.e., levels of connectedness, joint attention, and reciprocity) during the negative discussion. These studies suggest that physiological linkage indicates the quality of the relationship or interaction in various ways.

However, existing findings on the relationship between physiological linkage and interpersonal dynamics are mixed and complicated. The examples above suggest that higher linkage indicates better relationship quality; however, some researchers also report that higher linkage is associated with adverse outcomes, depending on contexts. For example, Levenson and Gottman [41] reported that higher physiological linkage during couples' discussions on their problem area was associated with lower marital satisfaction; Wilson et al [72] found a negative association between physiological linkage and couples' momentary closeness. These discrepancies may arise from differences in experimental conditions (e.g., positive or negative interactions), dyadic configurations (e.g., romantic partners, mother-child, or therapist-client), types of ANS activities associated with the measured physiologic modalities (i.e., sympathetic nervous system [SNS] or parasympathetic nervous system [PNS] activities), direction of linkage (i.e., in-phase or anti-phase) and the methods used to quantify linkage (e.g., Pearson's correlation, wavelet coherence, or differential equations). Such differences make a direct comparison and interpretation of research in this area challenging.

Prior reviews have examined physiological linkage broadly and comprehensively [55] or focused on particular types of dyadic configurations [39, 18, 21, 73]. Gordon and Bartsch also reviewed empirical studies

and discussed their methodological variability [29]. Recently, some effort has been made to consolidate the existing evidence around the associations between physiological linkage and relationship dynamics through the lens of ANS activities [66, 49]. For example, Timmons et al. [66] reviewed papers focusing on the associations between physiological linkage and the relationship quality variables (i.e., relationship satisfaction, empathy, physical proximity, emotional experience, and others) in couples. They found partial evidence that higher linkage in SNS activities indicates a lower relationship quality. Mayo et al. [49] conducted a meta-analysis to examine the association of physiological linkage to relationship quality and performance across different types of dyads. They found that linkage in SNS activities was positively associated with relationship quality, while linkage in PNS activities was negatively associated. These papers provide a good overview of the current studies in this field from the perspective of autonomic nervous system activities. However, the two reviews reached somewhat conflicting conclusions, both suggesting that associations between physiological linkage and relationship dynamics may vary depending on experimental conditions or the direction of linkage. These reviews also examined a relatively limited range of contextual factors that may moderate these associations. Thus, a systematic analysis is needed to examine how these associations vary across interaction contexts, autonomic nervous system branches, linkage directions, and quantification methods.

To close this gap, this work aims to systematically review papers using meta-analysis to examine the relationship between physiological linkage and relationship/interaction quality. This meta-analysis addresses the following research questions:

- **(RQ1)** What is the direction and magnitude of the association between physiological linkage and interpersonal, relational, socioemotional, behavioral, or performance-related outcomes?
- **(RQ2)** Does this association differ across physiological systems, particularly between SNS, PNS, and measures reflecting combined ANS activity?
- **(RQ3)** Under what conditions does the association between physiological linkage and interpersonal outcomes vary? Specifically, is the association moderated by dyad or group type, interaction context, outcome type, the direction of linkage captured by the measure, or the method used to quantify linkage?

More broadly, this review examines what existing evidence can tell us about when, for whom, and through which physiological signals and linkage patterns greater physiological linkage relates to more favorable or less favorable interpersonal outcomes. By examining findings across contexts, physiological modalities, populations, outcomes, and analytical methods, we assess how consistently the evidence supports these distinctions and where uncertainty remains. To the best of our knowledge, the present study is the first to jointly examine such a broad range of contextual and methodological moderators of the associations between physiological linkage and relationship dynamics. This synthesis highlights considerations for selecting and interpreting linkage measures and identifies questions that future studies need to address.

# 2 Methods

In this section, we present the methodology of our meta-analysis, including the search strategy, study selection, data extraction, and analytical approaches.

## 2.1 Literature search

We used PubMed, PsycInfo, and PsycArticles as target academic databases and entered the following search terms: “physiological synchrony” OR “ interpersonal physiology” OR “physiological linkage” OR “physi-

ological coherence" OR "physiological covariation" OR "physiological coregulation" OR "dyadic physiology." We selected these terms by referring to Palumbo et al.'s [55] search terms as well as the terminologies used in their identified papers. Because some systematic reviews have been conducted in this research area, we also searched cited articles in the identified review papers [55, 39, 21, 21, 73, 66, 49] using Web of Science. This search was last updated on June 11th, 2025.

## 2.2 Study selection

The first author screened all records based on their titles and abstracts. Studies were eligible if they empirically examined the association between physiological linkage among two or more individuals and an interpersonal, relational, socioemotional, behavioral, or performance-related outcome. Eligible outcomes included relationship or interaction quality, affective experience, empathy, responsiveness, interpersonal perceptions, behavioral coordination, cooperation, group functioning, and individual or collective task performance within an interpersonal context. Records for which eligibility could not be determined from the title and abstract were retained for full-text review.

We excluded studies that: (1) were not available in full text; (2) were not written in English; (3) were not published in a peer-reviewed journal or conference proceeding; (4) were reviews or other non-empirical publications; (5) were single-case studies involving only one dyad, group, or other interpersonal unit; (6) did not empirically assess physiological linkage using physiological data collected from human participants; (7) did not include a measure of autonomic physiology; (8) did not examine an eligible interpersonal, relational, socioemotional, behavioral, or performance-related outcome; or (9) examined the relationship between physiological linkage and an eligible outcome only qualitatively, without a quantitative assessment of the association. The primary author screened the papers, and whenever uncertainty arose over the eligibility of a given article, expert consensus was obtained by the co-authors to resolve ambiguity.

For inclusion in the quantitative meta-analysis, studies were additionally required to report a correlation coefficient or sufficient statistical information to calculate or convert the reported result into a correlation-based effect size. When an otherwise eligible study did not provide sufficient information, we contacted the authors to request the relevant effect size or supplementary statistics. Studies were excluded from the quantitative synthesis when no association measure could be obtained from the publication, supplementary materials, open-source datasets, or correspondence with the authors.

## 2.3 Data extraction

The first author extracted the following information from each included study: (1) the type of relationship or interpersonal group examined; (2) the sample size used in the relevant analysis; (3) the physiological measures and corresponding autonomic nervous system activity; (4) the experimental or interaction conditions; (5) the interpersonal, relational, socioemotional, behavioral, or performance-related outcomes; (6) the methods used to quantify physiological linkage; (7) the polarity of linkage captured by each method; (8) the analytic direction of the association between physiological linkage and the relevant outcome; and (9) the reported effect sizes and statistical information required to convert them into correlation-based effect sizes.

We coded the polarity of each linkage measure because statistical methods differ in how they represent the direction of physiological linkage. Some methods quantify only in-phase linkage, some combine in-phase and anti-phase linkage into a single nonnegative measure of linkage magnitude, and others produce signed or bipolar values that distinguish in-phase from anti-phase linkage.

We also coded the analytic direction of each reported association. Effect sizes were classified as: (1) bivariate or otherwise symmetric associations; (2) models in which physiological linkage was predicted by an interpersonal, relational, socioemotional, behavioral, or performance-related variable; or (3) models in

which such a variable was predicted by physiological linkage. This classification reflects the specification of the statistical model and does not necessarily imply causal direction.

Finally, we coded the direction of each outcome measure to ensure that effect sizes had a consistent interpretation. For some outcomes, higher values represented less favorable interpersonal or socioemotional functioning, such as greater negative affect or psychological unavailability. For these outcomes, we reversed the sign of the effect size so that positive effect sizes consistently indicated that greater physiological linkage was associated with more favorable outcomes, whereas negative effect sizes indicated an association with less favorable outcomes.

## 2.4 Meta-analysis

All meta-analyses were conducted on Fisher-transformed correlation coefficients, with estimates back-transformed to $r$ for interpretation. Unadjusted and covariate-adjusted effect sizes were analyzed separately. Because individual studies frequently contributed multiple effect sizes, we accounted for their dependence by constructing an approximate within-study sampling variance-covariance matrix, assuming a sampling-error correlation of $\rho = .60$, as suggested by Assink and Wibbelink [2]. For the study-level forest plots and funnel plots, multiple effect sizes from the same study were aggregated into a single study-level estimate on the Fisher's $z$ scale using `metafor::aggregate()`. Study-level aggregation was used for visualization and small-study-effect diagnostics only. We then fit multilevel random-effects models using restricted maximum likelihood estimation, with effect sizes nested within studies. To further account for dependence among effects from the same study, statistical inference was based on study-clustered robust (sandwich-type) variance estimation with a CR2 small-sample correction.

We first estimated the overall association between physiological linkage and interpersonal outcomes and then examined moderators using multilevel meta-regression. Moderators included ANS branch, outcome category, linkage type, and several aspects of interaction context. Outcome categories included relationship quality, empathy or understanding, affect, interaction behavior, collective performance, clinical alliance, and individual relational characteristics.

We described interaction context along three dimensions: emotional tone, interaction format, and the presence of specific interaction features. Affective context captured emotional tone, with categories of positive, negative, neutral, mixed, or not explicitly valenced. Neutral referred to contexts explicitly described or designed as emotionally neutral; mixed referred to contexts that included multiple emotional valences; and not explicitly valenced referred to contexts without a clearly specified emotional valence. Interaction format captured how participants engaged with each other, with categories of structured discussion, unstructured interaction, goal-oriented joint activity, clinical or helping interaction, and shared stimulus or passive observation. We also used three separate binary labels to indicate whether the interaction was stressful, helping/supportive, or goal-oriented. These features could overlap and occur across different formats. For example, a structured discussion could involve both stress and support, while a joint task could be goal-oriented with or without stress. The binary labels allowed us to examine these features across formats without creating a separate category for every combination.

We additionally examined three characteristics of the linkage quantification approach: whether the method explicitly considered non-zero temporal lags between participants' physiological signals, whether it explicitly modeled or adjusted for within-person temporal dependency such as serial autocorrelation, and the broader methodological family used to quantify linkage. We grouped linkage methods into five categories based on their underlying computational strategy rather than method name alone. Correlation-based approaches measure how partners' physiological signals vary together, either simultaneously or with a time lag. Coherence-based approaches measure the correspondence between signals at particular frequencies or time–frequency regions. Pattern-similarity approaches compare the shapes or sequences of physiological responses. Local-concordance approaches assess whether partners' signals show matching changes, such

as increasing or decreasing together, within short time intervals. Model-based coupling approaches estimate associations between partners' physiological responses through parameters in statistical or dynamical models.

Moderators were initially examined separately to limit model complexity. Given our primary interest in differences between autonomic systems, targeted analyses compared effects classified as PNS versus SNS and tested whether this difference varied as a function of stressful, helping, and goal-oriented contexts. Cell-specific pooled estimates and the number of independent studies contributing to each combination were examined alongside interaction tests.

Sensitivity analyses evaluated the robustness of the findings to assumptions about dependent effect sizes by repeating models with within-study correlations of $\rho = .30$ and $\rho = .90$. We additionally repeated the unadjusted analysis using Pearson correlations only, excluding Spearman correlations, and repeated the adjusted analysis after excluding effects derived from mixed- or multilevel-model statistics. All analyses were conducted in R using the `metafor`, `clubSandwich`, and `tidyverse` packages.

# 3 Results

In this section, we present the study characteristics and the analysis results.

## 3.1 Study characteristics

The final extraction dataset contained 392 eligible effect-size records from 50 studies (Figure 2). Of these, 214 uncorrected effect sizes from 34 studies provided sufficient information for the primary meta-analysis. The corrected-effect analysis included 142 effect sizes from 15 studies. Among the corrected effects, 112 were derived from ordinary regression models and 30 were approximate partial correlations derived from mixed- or multilevel-model results.

The uncorrected effect sizes ranged from $r = -.77$ to $r = .65$ ($M = -.03$, $SD = .24$), with a median of $r = -.04$. Corrected effect sizes ranged from $r = -.62$ to $r = .54$ ($M = .00$, $SD = .19$), with a median of $r = -.01$.

The uncorrected dataset included 31 effects reflecting predominantly PNS activity, 36 reflecting predominantly SNS activity, and 147 measures for which the two autonomic branches were not distinguished. The most frequently represented outcome categories were interaction behavior ($k = 88$ effects from 9 studies), relationship quality ($k = 51$ from 14 studies), and collective performance ($k = 33$ from 8 studies), followed by affect, empathy or understanding, clinical alliance, and individual relational characteristics. With respect to the directionality represented by the linkage measure, 102 effects were coded as bipolar, 62 as in-phase, 44 as pooling different directions of linkage, and 6 as anti-phase.

## 3.2 Direction and magnitude of the association (RQ1)

To address RQ1, we first estimated the overall association between physiological linkage and interpersonal outcomes. The multilevel meta-analysis of 214 uncorrected effects from 34 studies yielded a small positive pooled association, $r = .030$, 95% CI $[-.024, .084]$ (Figure 3), which was not statistically different from zero, $p = .262$. The prediction interval was wide, $[-.307, .360]$, indicating that associations in individual settings could plausibly vary in both positive and negative directions. Significant heterogeneity was observed across the included effects, $Q(213) = 646.65$, $p < .001$.

The corresponding analysis of 142 corrected effects from 15 studies also yielded a small positive pooled association, $r = .066$, 95% CI $[-.014, .145]$, $p = .098$ (Figure 4). The prediction interval again encompassed both negative and positive associations, $[-.204, .327]$, and residual heterogeneity was significant, $Q(141) = 383.80$, $p < .001$.

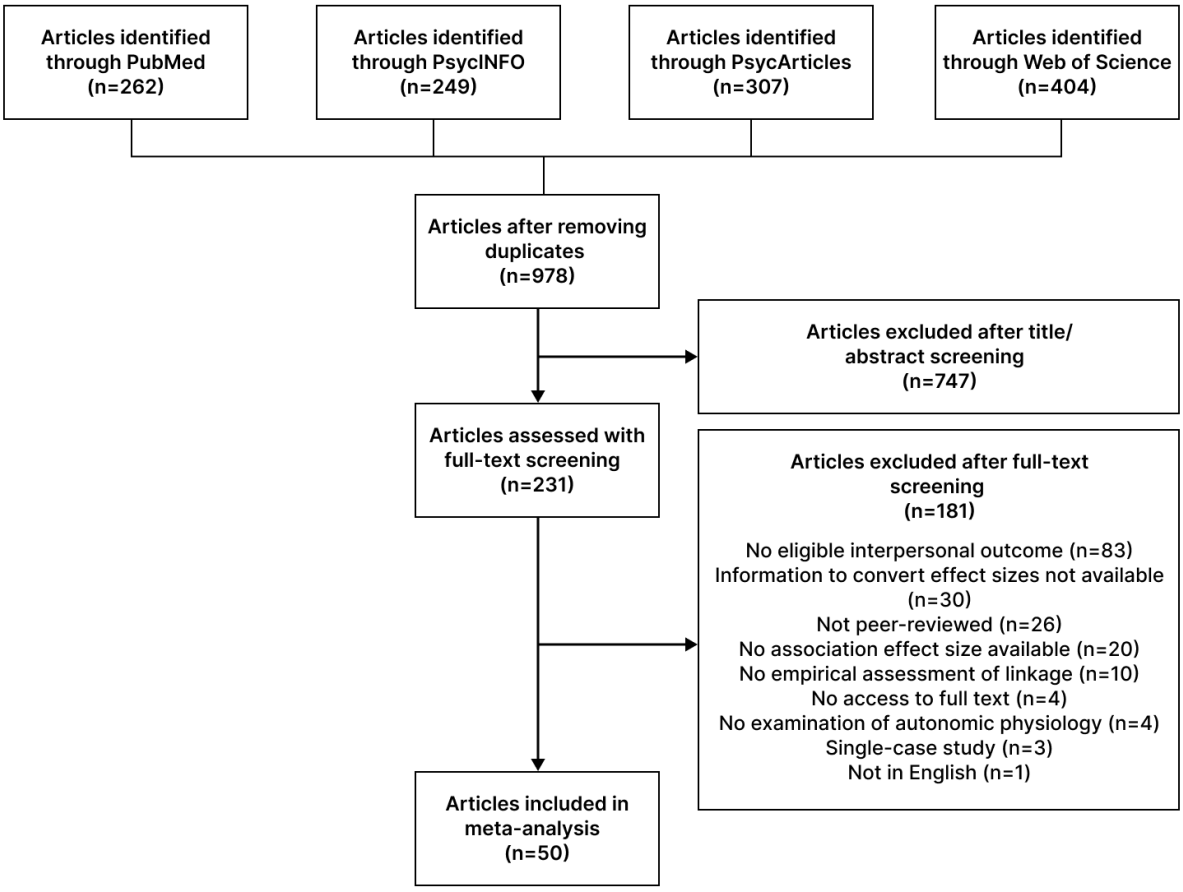



Figure 2: A flow diagram of article selection process.

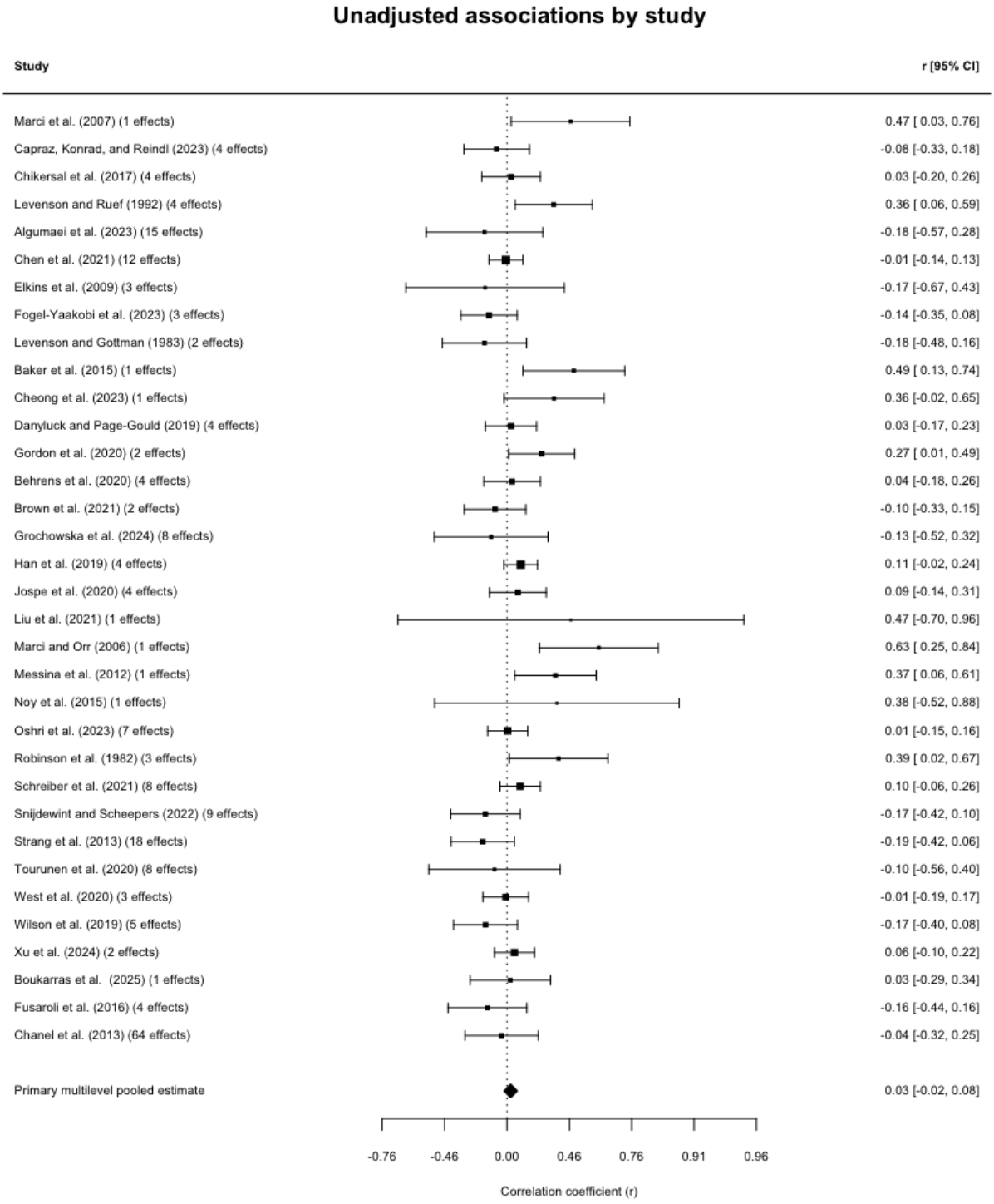



Figure 3: Forest plot for unadjusted effect sizes. Study-level estimates were obtained by aggregating dependent effect sizes within each study.

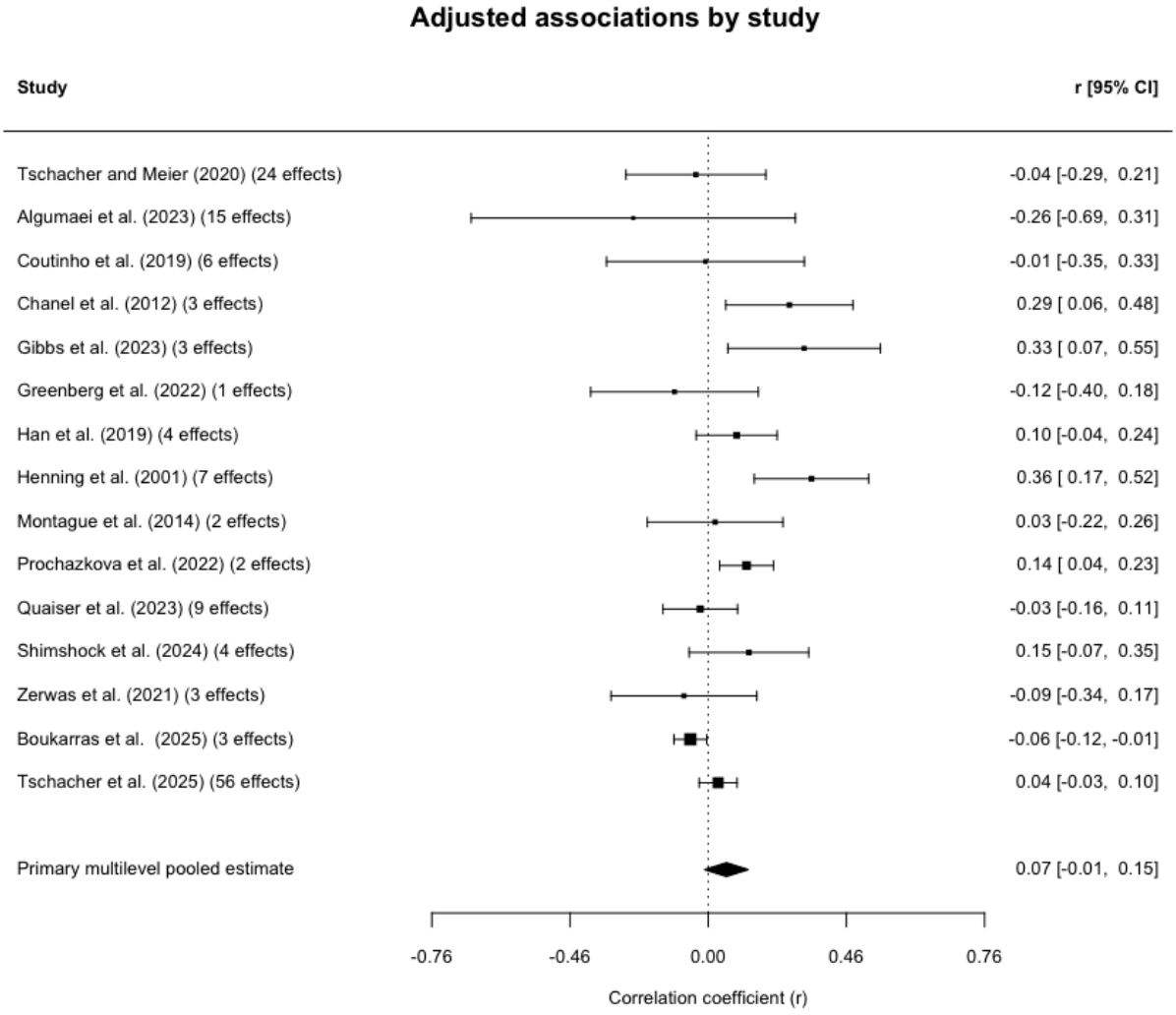



Figure 4: Forest plot for adjusted effect sizes. Study-level estimates were obtained by aggregating dependent effect sizes within each study.

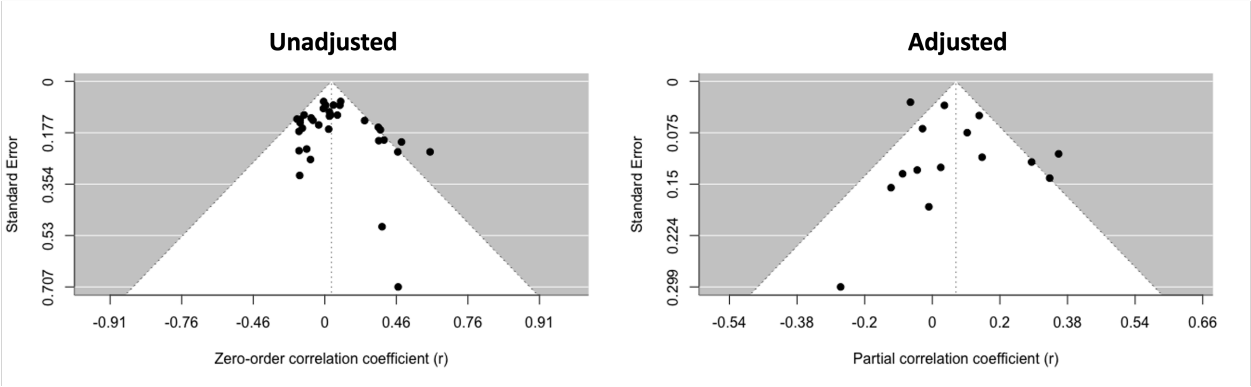



Figure 5: Funnel plots for unadjusted and adjusted effect sizes.

These findings therefore did not indicate a consistent overall positive or negative association between physiological linkage and interpersonal outcomes. Instead, the substantial heterogeneity motivated examination of whether the association differed according to physiological and methodological characteristics and interaction contexts.

Sensitivity analyses were consistent with the primary findings. Restricting the uncorrected analysis to Pearson correlations yielded $r = .039$, 95% CI $[-.017, .095]$. Likewise, excluding mixed- or multilevel-model-derived effects from the corrected analysis yielded $r = .078$, 95% CI $[-.115, .265]$.

We examined possible small-study effects using funnel plots (Figure 5) and Egger's test. Egger's test did not provide statistically significant evidence of funnel-plot asymmetry.

### 3.3 Differences across autonomic nervous system branches (RQ2)

To examine whether the association differed across physiological systems, we first compared effects classified as PNS, SNS, or measures that did not distinguish between autonomic branches. In the full uncorrected dataset, ANS classification did not significantly moderate the association, $F(2, 7.54) = 1.86$, $p = .221$. The corresponding moderator effect was also nonsignificant among corrected effect sizes, $F(2, 1.08) = 0.38$, $p = .750$.

Because measures classified as "no distinction" do not isolate either autonomic branch, we additionally conducted a targeted comparison restricted to PNS and SNS measures. This analysis included 67 uncorrected effects from 21 studies: 31 PNS effects from 8 studies and 36 SNS effects from 14 studies. In this subset, the association differed significantly between PNS and SNS measures, $F(1, 7.20) = 7.37$, $p = .029$. The pooled association for PNS measures was slightly negative ($r \approx -.06$), whereas the corresponding estimate for SNS measures was positive ($r \approx .14$). The estimated SNS–PNS difference on the Fisher's $z$ scale was $0.194$, 95% CI $[0.026, 0.362]$.

This difference was not replicated in the corrected-effect analysis, $F(1, 1.03) = 2.69$, $p = .344$. However, the corrected PNS–SNS comparison was highly imbalanced: only four PNS effects from two studies were available, compared with 74 SNS effects from ten studies. The corrected comparison should therefore be interpreted cautiously.

### 3.4 Moderator analyses (RQ3)

We next examined whether the association between physiological linkage and interpersonal outcomes varied as a function of interaction context, outcome type, linkage direction, and the method used to quantify linkage, separately for unadjusted and adjusted effect sizes (Figure 6). Models were estimated separately for

each moderator to avoid simultaneously fitting a large number of sparsely represented categorical predictors, with additional targeted interaction analyses.

**Relationship type.** Relationship type did not significantly moderate the association between physiological linkage and interpersonal outcomes in the uncorrected dataset, $F(4, 9.31) = 2.65$, $p = .101$. The uncorrected data included couples, parent–child dyads, strangers, teammates, and therapist–client dyads. Although the therapist–client coefficient differed significantly from the couples reference category, the overall omnibus moderator test was not significant, and this individual contrast was therefore not interpreted as evidence of systematic moderation by relationship type. Relationship type was also not a significant moderator in the corrected dataset, $F(5, 0.01) = 0.02$, $p = .996$. However, the corrected analysis was substantially sparser, with several relationship types represented by only one or two independent studies, limiting inference regarding differences across relationship types.

**Interaction context.** None of the prespecified binary context dimensions significantly moderated the uncorrected association. Effect sizes did not differ significantly between stressful and non-stressful contexts, $F(1, 17.75) = 0.26$, $p = .618$, or between goal-oriented and non-goal-oriented contexts, $F(1, 17.98) = 1.98$, $p = .177$. There was some indication of stronger associations in helping contexts, but the moderator test did not reach statistical significance, $F(1, 5.28) = 4.07$, $p = .097$. Likewise, neither the broader affective context classification, $F(4, 1.67) = 0.67$, $p = .680$, nor interaction format, $F(4, 2.82) = 0.89$, $p = .564$, significantly accounted for variation in the uncorrected effects.

The corrected analyses yielded a similar overall pattern. Stressful context showed the largest difference, but did not reach statistical significance, $F(1, 3.42) = 7.46$, $p = .062$. Helping context ($p = .136$), goal-oriented context ($p = .191$), affective context ($p = .826$), and interaction format ($p = .273$) were also not significant moderators. These corrected analyses were based on substantially fewer independent studies, particularly within several context categories.

**Outcome type.** Associations did not significantly differ across outcome categories in the uncorrected analysis, $F(6, 4.13) = 2.49$, $p = .194$. Thus, there was no clear evidence that physiological linkage was more consistently associated with affective, relational, behavioral, empathic, clinical-alliance, or performance-related outcomes. The corrected dataset was too sparse across outcome categories to provide a stable omnibus CR2 test.

**Linkage direction.** The type of linkage captured by the measure also did not significantly moderate the uncorrected association. Effects classified as in-phase, anti-phase, bipolar, or pooling linkage directions did not differ significantly overall, $F(3, 1.94) = 0.95$, $p = .553$. The corresponding corrected analysis similarly showed no evidence of differences among the available linkage-direction categories, $F(2, 0.83) = 0.15$, $p = .881$.

**Linkage quantification characteristics.** We examined whether associations varied according to whether the linkage quantification method considered non-zero temporal lags between participants, explicitly adjusted for within-person temporal dependency, and the broader linkage method family. In the uncorrected dataset, none of these characteristics significantly moderated the association: lag consideration, $F(1, 11.28) = 1.40$, $p = .262$; temporal-dependency adjustment, $F(1, 6.26) = 0.69$, $p = .437$; and linkage method family, $F(4, 1.28) = 6.26$, $p = .233$. The corrected analyses likewise showed no significant moderation by lag consideration, $F(1, 2.52) = 0.70$, $p = .473$, or temporal-dependency adjustment, $F(1, 3.11) = 6.54$, $p = .081$. However, only one study contributed corrected effects from methods coded

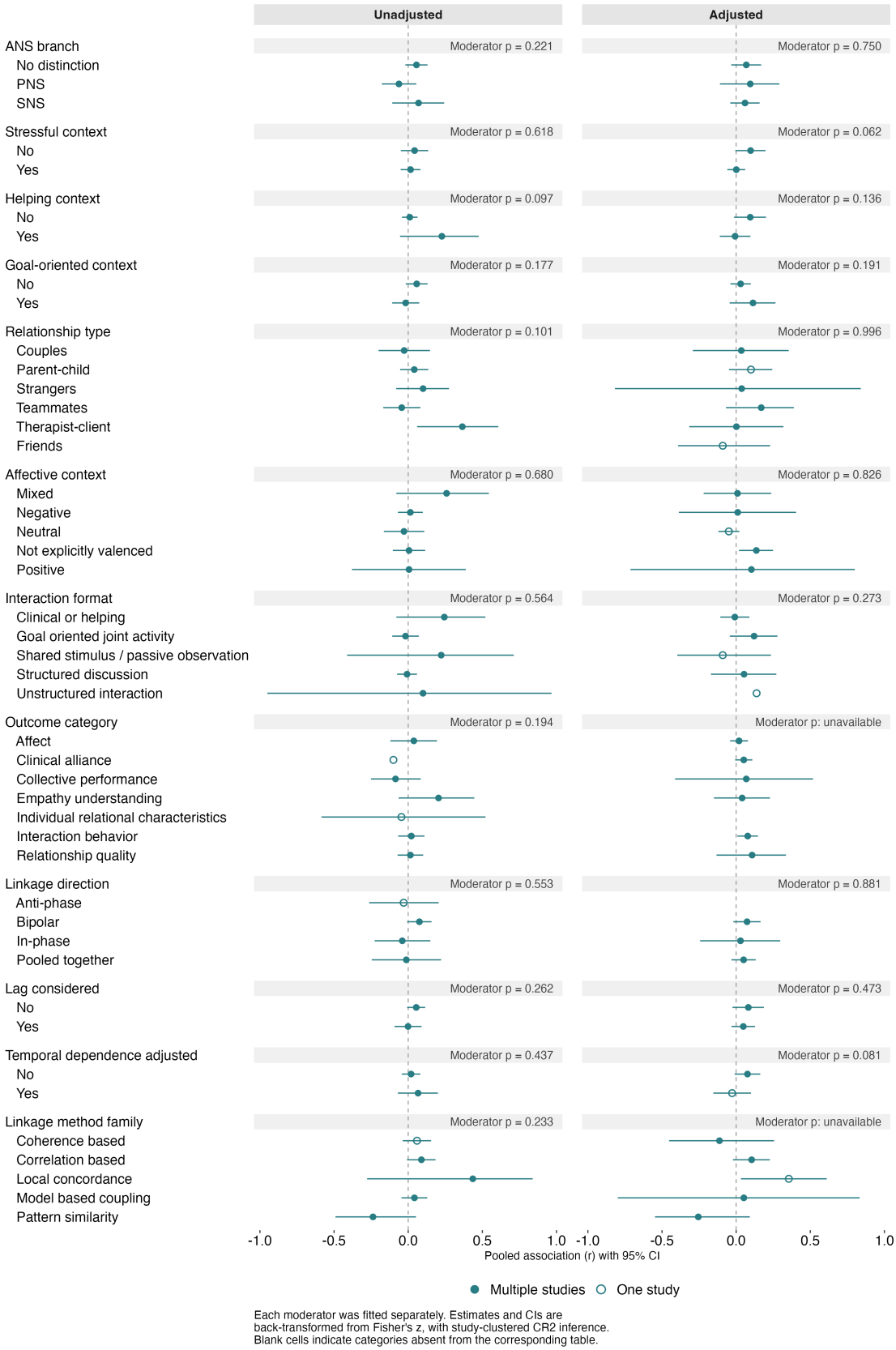



Figure 6: Forest plot for moderator analyses.

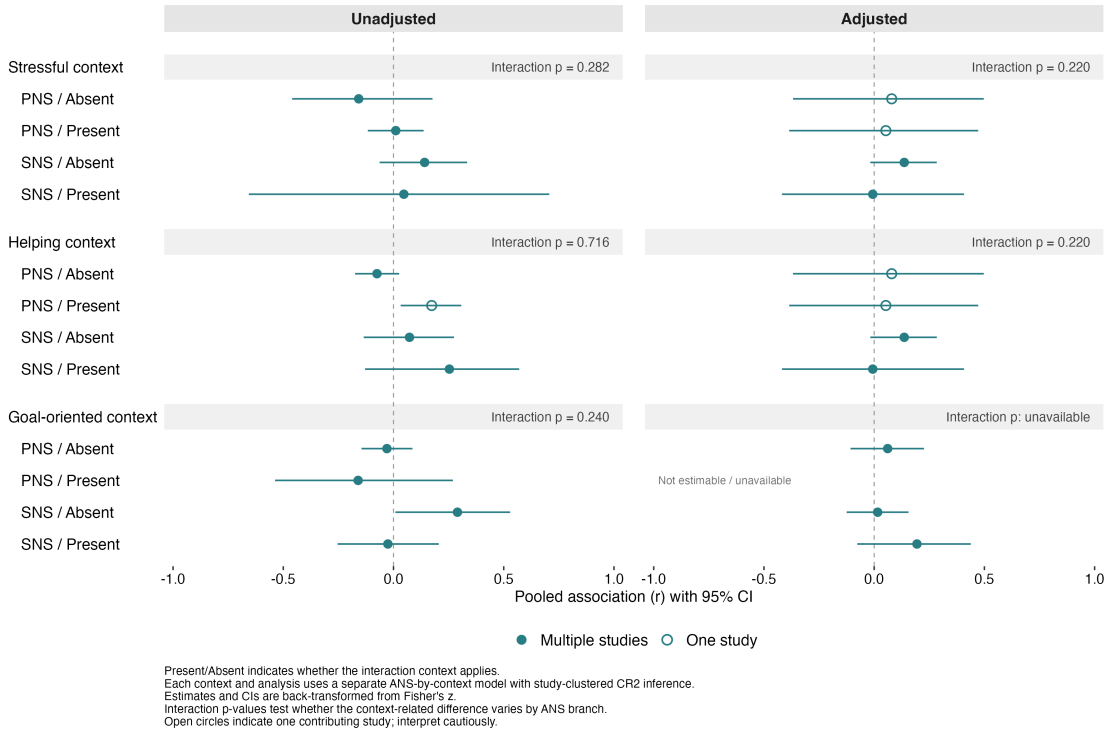



Figure 7: Forest plot for interaction effects.

as adjusting for temporal dependency. The omnibus test for linkage method family could not be reliably estimated in the corrected dataset because several method families were represented by very few independent studies.

**Interactions between physiological system, and context.** We additionally examined whether contextual effects depended on the ANS branch represented by the physiological measure (Figure 7). None of the targeted ANS-by-context interactions was statistically significant: ANS $\times$ stressful context ($p = .282$), ANS $\times$ helping context ($p = .716$), or ANS $\times$ goal-oriented context ($p = .240$). Corrected analyses likewise provided no clear evidence of ANS-by-context interactions and were characterized by sparse cells.

# 4 Discussion

This review examined how physiological linkage relates to interpersonal outcomes and how these associations vary across methods and study characteristics. We found a small average association alongside substantial variation across effects. The restricted comparison of SNS and PNS measures suggested a difference between autonomic branches, but most moderator analyses did not clearly explain the variation. Together, these findings highlight the need to specify what physiological linkage captures before interpreting its relationship with interpersonal outcomes. Our review contributes to this effort by bringing together the quantitative evidence, documenting methodological differences, and providing a literature database for future research.

### 4.1 Physiological linkage and interpersonal outcomes

The overall findings do not support treating greater physiological linkage as a general indicator of better interpersonal outcomes. Both unadjusted and adjusted analyses showed small average associations, and neither reached statistical significance. At the same time, substantial heterogeneity and a wide prediction interval suggest that this average does not describe every setting well. The central question for future research is therefore when, how, and for whom linkage relates to particular outcomes.

Previous reviews have emphasized that linkage can accompany both positive interpersonal experiences and shared distress [66, 29]. For example, coordinating with another person may support mutual understanding, but sharing and sustaining each other's stress may have different implications. Our findings are consistent with these context-sensitive accounts, although our moderator analyses did not establish which contexts explain the differences. They support caution about assigning a single positive or negative meaning to physiological linkage.

The outcome of interest also matters. Our review included different relationship types and outcomes, including satisfaction, empathy, interpersonal behavior, and performance. These outcomes capture different aspects of an interaction, so the same physiological pattern may not relate to all of them in the same way. Future studies should state which interpersonal process they expect linkage to reflect and why that process should relate to their chosen outcome. This would make findings easier to interpret and help later reviews compare studies that address similar questions.

### 4.2 The role of ANS activity

The SNS–PNS comparison suggests that distinguishing autonomic branches may help clarify the meaning of linkage. In the restricted unadjusted analysis, SNS measures showed a significantly more positive association with interpersonal outcomes than PNS measures. The estimated correlations were approximately $.137$ for SNS and $-.057$ for PNS. These directions agree with an earlier meta-analysis, although our PNS estimate was smaller in magnitude [49]. However, the overall test including the "No distinction" category was nonsignificant. This category accounted for the vast majority of effect sizes, and its estimated association fell between those for SNS and PNS linkage, which may have contributed to the lack of clear differentiation across the three categories. The SNS–PNS comparison was also nonsignificant in the adjusted subset. Nevertheless, the potential branch-specific pattern, consistent with prior literature, highlights the value of reporting and analyzing SNS and PNS linkage separately, where possible. We therefore interpret this finding as motivation for further investigation of branch-specific associations, rather than as evidence that SNS linkage is generally beneficial and PNS linkage is harmful.

Differences across reviews also point to the importance of the question and evidence each review covers. Our review differs from the earlier meta-analysis in its included evidence, physiological classifications, and handling of dependent and adjusted effects, which may contribute to differences in the estimates. A review of romantic couples found partial support for a relationship between SNS or hypothalamic–pituitary–adrenal (HPA) coactivation and poorer relationship functioning [66]. Its focus on couples and inclusion of cortisol findings limit direct comparison with our broader synthesis. Taken together, these reviews motivate more specific hypotheses about the physiological system, interaction, and outcome, rather than a fixed interpretation of linkage within each branch.

The physiology itself provides a reason for this specificity. SNS and PNS activity do not always change in opposite directions, and different combinations of activity can produce similar changes in heart rate or interbeat interval [58]. Even within HRV, each measure differs in the variation they capture and in how breathing affects its interpretation [58]. Two people who show similar changes in a cardiac measure may therefore differ in their underlying autonomic responses. Combining several modalities into one linkage score can further make it difficult to identify which physiological process relates to the outcome.

Researchers should choose physiological measures that fit their hypotheses and explain what those measures allow them to infer. When a question concerns a particular autonomic branch, branch-specific measures and separate analyses of relevant modalities would provide clearer evidence. Researchers should also consider the physiological state in which linkage occurs: partners may show positive linkage while both increase or both decrease their responses. Measuring how their individual responses change alongside their linkage can help distinguish these possibilities.

Context remains an important question for this work. Our ANS-by-context analyses did not identify significant interactions, although some estimates differed descriptively. For example, in the unadjusted analysis, SNS linkage showed a less positive association with interpersonal outcomes in stressful than in non-stressful contexts, whereas PNS linkage showed a near-zero association in stressful contexts and a negative association in non-stressful contexts. These descriptive differences suggest patterns for future studies to examine, but the wide confidence intervals and limited evidence within each combination prevent firm conclusions about whether context changes the association differently for SNS and PNS linkage. Some researchers have also suggested that linkage during stressful interactions may reflect partners amplifying each other's stress [66]. Our findings do not establish that linkage in stressful contexts is associated with less favorable interpersonal outcomes, and our analyses cannot directly test mutual stress escalation. This interpretation therefore requires caution. Future studies that measure both autonomic branches alongside changes in each partner's stress across well-defined contexts could test this possibility more directly and clarify when linkage accompanies support, shared stress, or other interpersonal processes.

### 4.3 Methodological heterogenity

Our review shows that physiological linkage covers a wide range of measurements and analytical choices. Studies differ in preprocessing, window size, lag, directionality, and how they treat in-phase and anti-phase patterns. These choices define which aspects of coordination a measure captures. Although our moderator analyses could not establish their contributions to differences in linkage–outcome associations, researchers still need to justify them in relation to the process they want to understand.

**Choose a meaningful timescale.** The time window should fit both the physiological signal and the interpersonal process of interest [47]. A whole-interaction score may suit questions about sustained coordination, while shorter windows may better capture responses during specific events, such as offering support or responding to disagreement. Researchers should explain why their window captures the relevant process and, when appropriate, check whether their conclusions hold across other reasonable window sizes. Behavioral coding can help identify meaningful episodes independently of the physiological linkage estimate.

Attention to timing also allows researchers to study changes within an interaction. Theoretical work suggests that people move between coordination and independence rather than remain continuously synchronized [48]. A single average may miss these transitions. Examining when linkage increases, decreases, or changes direction may therefore reveal relationships that a whole-task score cannot capture. This is a direction for future research, rather than a process that our meta-analysis directly tested.

**Define the form of linkage.** Researchers should specify whether they expect simultaneous changes, delayed responses, shared slow trends, or another form of coordination. They should also explain whether the roles of the two partners matter and whether in-phase and anti-phase patterns have different meanings for their question. These distinctions help connect the statistical measure to a psychological interpretation. Preprocessing should follow the same logic: removing trends or adjusting response scales changes which features some methods retain, so researchers should make these decisions explicit.

Method comparisons support this need to match the analysis to the expected pattern. Different correlation, wavelet, recurrence, and distance approaches capture different aspects of physiological coordi-

nation [20]. No single method necessarily answers every linkage question. Where a hypothesis involves changing response delays, methods that allow lag to vary over time may be useful, provided that researchers constrain the alignment and evaluate it against suitable null comparisons. More broadly, researchers should justify their parameter choices and show how those choices affect the conclusions they draw about interpersonal outcomes.

How a measure treats in-phase and anti-phase linkage also changes the meaning of its association with relationship outcomes. An in-phase-only measure asks whether stronger in-phase linkage is associated with more favorable outcomes. A pooled measure combines in-phase and anti-phase linkage, asking whether stronger linkage regardless of direction is associated with more favorable outcomes, without distinguishing their separate contributions. A bipolar measure instead places anti-phase and in-phase linkage at opposite ends of a signed scale. A positive association therefore indicates more favorable outcomes as linkage shifts toward the in-phase end, but does not establish that in-phase linkage is beneficial and anti-phase linkage is harmful. For example, if observations are mostly in-phase, the association may primarily reflect differences in the strength of in-phase linkage. Conversely, a near-zero linear association could mask a pattern in which both strong in-phase and strong anti-phase linkage accompany more favorable outcomes than weak linkage.

Although we found no significant differences across these categories, their different meanings remain important for interpretation. Future studies could examine in-phase and anti-phase linkage separately and test whether their associations with outcomes are nonlinear. We wanted to compare their separate associations in this review, but studies often did not report separate estimates. Reporting both would help future reviews assess when each form of linkage relates to interpersonal outcomes.

**Evaluate alternative explanations and measurement quality.** An interpretable linkage estimate also requires attention to autocorrelation, shared events, and artifacts. Partners may respond to the same event without directly influencing each other, and differences in response timing can produce lagged as well as simultaneous linkage. Surrogate procedures, such as random-pair or time-shift comparisons, can help evaluate specific alternative explanations when researchers choose a null model that fits the question [29]. These checks strengthen the interpretation of the linkage measure; analyses relating linkage to interpersonal outcomes address a separate question.

Researchers should give similar attention to outcome measurement. The studies in our review used established questionnaires, adapted item sets, and custom measures. Clear reporting of reliability, validity, and the reasons for choosing a measure would help readers judge what an association means. Improving the measurement of both linkage and outcomes is essential for separating weak associations from uncertainty introduced by measurement.

## 4.4 Limitations and future research

The main limitation is the uneven evidence available for specific comparisons. The unadjusted analysis included 214 effects from 34 studies, but some moderator categories and combinations contained very few independent studies. Our models accounted for dependent effects, yet the large effect-size count does not overcome these gaps. In addition, methods often varied alongside populations, tasks, and outcomes, making it difficult to separate their contributions. The restricted SNS–PNS finding also arose within a broader set of exploratory analyses and needs further testing.

Our broad scope helped us describe the field, but it required categories that simplify differences between studies. Binary context variables and groups for methods, interaction formats, and outcomes may hide meaningful differences within each category. Likewise, aligning the direction of outcome effects does not make the underlying constructs equivalent. Future reviews with more evidence could examine narrower questions or use more detailed classifications.

Adjusted and unadjusted effects also represent different associations. Studies controlled for different variables, and some effect-size conversions from mixed models required approximations. We analyzed adjusted and unadjusted effects separately, but differences in the evidence available for each subset still limit comparisons. The Pearson-only analysis and the analysis excluding adjusted effects derived from mixed models both retained small, nonsignificant average estimates. These sensitivity analyses support the overall interpretation, although uncertainty remains and they do not establish the robustness of every moderator finding.

Missing evidence further limits coverage. We excluded 30 studies because we lacked the information needed to calculate or convert effect sizes, and the authors either did not respond to our requests or could no longer locate the relevant information. Including these studies could change the estimates, heterogeneity, or precision in either direction. Publication bias and selective reporting are additional concerns. Although neither Egger-type test reached statistical significance, these tests do not rule out reporting bias. We also did not systematically track whether each study reported all nonsignificant associations, which limits our assessment of selective reporting.

A useful next step is to compare methods within the same datasets. Applying several physiological measures and analysis approaches to common recordings would help separate methodological differences from differences in participants and tasks. These comparisons should evaluate how choices such as window size, lag, and preprocessing affect both the linkage estimate and its relationship with outcomes. Clear hypotheses, transparent parameter choices, and reporting of all relevant results would make these comparisons more useful for the field.

The literature database provides a starting point for this work. We open-source the literature and effect-size database we collected, which helps other researchers check our decisions, test alternative groupings, and add new evidence. Together with more focused comparisons, this could help the field move toward a clearer understanding of which forms of physiological linkage matter for which interpersonal outcomes.

# 5 Conclusion

This work contributes a systematic review and meta-analysis of the association between physiological linkage and interpersonal outcomes, a quantitative examination of methodological and theoretically relevant study characteristics, and an open-source literature database that supports replication and future updates. We found small, nonsignificant average associations alongside substantial variation across effects. The restricted unadjusted analysis suggested a more positive association for SNS than PNS measures, while most moderator analyses provided limited evidence to explain the variation. These findings do not support treating greater linkage as a general indicator of better interpersonal outcomes.

By documenting the range of physiological measures, contexts, and analysis methods across studies, this review identifies key choices that future research should consider. Researchers should connect these choices to clear hypotheses and compare methods within the same datasets to better understand which forms of linkage relate to which outcomes. Our open-source database makes the evidence accessible for researchers to examine coding decisions, test alternative classifications, and extend the synthesis with new studies.

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# A Appendix

Table 1: Extracted statistics for associations between physiological linkage and relational, affective, behavioral, or performance outcomes.

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; REC | In-phase | task performance | Pearson's correlation; U | $r = -0.253$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; DET | In-phase | task performance | Pearson's correlation; U | $r = -0.323$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; ADL | In-phase | task performance | Pearson's correlation; U | $r = -0.253$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; MDL | In-phase | task performance | Pearson's correlation; U | $r = -0.242$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | Cross-correlation | Bipolar | task performance | Pearson's correlation; U | $r = -0.075$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; REC | In-phase | task performance | Pearson's correlation; U | $r = 0.004$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; DET | In-phase | task performance | Pearson's correlation; U | $r = -0.213$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; ADL | In-phase | task performance | Pearson's correlation; U | $r = -0.09$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; MDL | In-phase | task performance | Pearson's correlation; U | $r = -0.113$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | Cross-correlation | Bipolar | task performance | Pearson's correlation; U | $r = 0.081$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; REC | In-phase | task performance | Pearson's correlation; U | $r = -0.195$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; DET | In-phase | task performance | Pearson's correlation; U | $r = -0.324$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; ADL | In-phase | task performance | Pearson's correlation; U | $r = -0.119$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; MDL | In-phase | task performance | Pearson's correlation; U | $r = -0.413$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | Cross-correlation | Bipolar | task performance | Pearson's correlation; U | $r = -0.095$ | — |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; REC | In-phase | self-reported performance | Multiple regression model; C | $t = -2.07$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; DET | In-phase | self-reported performance | Multiple regression model; C | $t = -0.76$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; ADL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.84$ | 8 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | MdRQA; MDL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.31$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | R-R interval | Cross-correlation | Bipolar | self-reported performance | Multiple regression model; C | $t = 0.84$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; REC | In-phase | self-reported performance | Multiple regression model; C | $t = 0.078$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; DET | In-phase | self-reported performance | Multiple regression model; C | $t = -1.11$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; ADL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.54$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | MdRQA; MDL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.72$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (RMSSD) | Cross-correlation | Bipolar | self-reported performance | Multiple regression model; C | $t = -0.04$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; REC | In-phase | self-reported performance | Multiple regression model; C | $t = -2.22$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; DET | In-phase | self-reported performance | Multiple regression model; C | $t = -1.54$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; ADL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.953$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | MdRQA; MDL | In-phase | self-reported performance | Multiple regression model; C | $t = -0.97$ | 8 |
| Algumaei et al. (2023) [1] | 14 triads | cooperative task | HRV (SDNN) | Cross-correlation | Bipolar | self-reported performance | Multiple regression model; C | $t = -0.84$ | 8 |
| Baker et al. (2015) [3] | 26 dyads | free play | EDA | Pearson's correlation | Bipolar | affective mutuality | Pearson's correlation + Hierarchical linear model; U | $r = 0.49$ | — |
| Behrens et al. (2020) [6] | 60 dyads | cooperative play face-to-face | HR | Cross-correlation | In-phase | team performance | Multilevel linear regression; U | $r = 0.0476$ | — |
| Behrens et al. (2020) [6] | 60 dyads | cooperative play face-blocking | HR | Cross-correlation | In-phase | team performance | Multilevel linear regression; U | $r = -0.0316$ | — |
| Behrens et al. (2020) [6] | 50 dyads | cooperative play face-to-face | SCL | Cross-correlation | In-phase | team performance | Multilevel linear regression; U | $r = 0.0763$ | — |
| Behrens et al. (2020) [6] | 50 dyads | cooperative play face-blocking | SCL | Cross-correlation | In-phase | team performance | Multilevel linear regression; U | $r = 0.1532$ | — |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | task performance | Mixed effects linear models; C | $t = 1.448$ | 560 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | task performance | Mixed effects linear models; C | $t = 0.347$ | 404 |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | perceived task performance | Mixed effects linear models; C | $t = -2.068$ | 1061 |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | perceived task easiness | Mixed effects linear models; C | $t = -2.002$ | — |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | perceived task pleasantness | Mixed effects linear models; C | $t = -1.469$ | — |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | perceived alignment | Mixed effects linear models; C | $t = -1.348$ | — |
| Boukarras et al. (2025) [8] | 40 dyads | cooperative task | IBI | Cross-correlation | Bipolar | perceived closeness | Mixed effects linear models; U | $\rho = 0.027$ | — |
| Brown et al. (2021) [9] | 54 dyads | discussion and listening of stressful events | PEP | Vector autoregression | Bipolar | empathic accuracy [Target: experiencer (talker)] | Dyadic regression modeling (mixed effects model); U | $r = -0.0937$ | — |
| Brown et al. (2021) [9] | 53 dyads | discussion and listening of stressful events | PEP | Vector autoregression | Bipolar | empathic accuracy [Target: listener] | Dyadic regression modeling (mixed effects model); U | $r = -0.0972$ | — |
| Capraz, Konrad, and Reindl (2023) [10] | 42 dyads | cooperative task (1st) | IBI | Cross-correlation | Bipolar | affect synchrony | Pearson's correlation; U | $r = -0.113$ | — |
| Capraz, Konrad, and Reindl (2023) [10] | 42 dyads | cooperative task (2nd) | IBI | Cross-correlation | Bipolar | affect synchrony | Pearson's correlation; U | $r = -0.055$ | — |
| Capraz, Konrad, and Reindl (2023) [10] | 42 dyads | cooperative task (1st) | RSA | Cross-correlation | Bipolar | affect synchrony | Pearson's correlation; U | $r = -0.149$ | — |
| Capraz, Konrad, and Reindl (2023) [10] | 42 dyads | cooperative task (2nd) | RSA | Cross-correlation | Bipolar | affect synchrony | Pearson's correlation; U | $r = -0.012$ | — |
| Chanel et al. (2012) [12] | 21 dyads | competitive and cooperative game play | EDA | Pearson's correlation | Bipolar | positive affect rating | Linear mixed model; C | $F = 9.34$ | 1, 60.18 |
| Chanel et al. (2012) [12] | 21 dyads | competitive and cooperative game play | R-R interval | Weighted coherence | Pooled | negative affect rating | Linear mixed model; C | $F = 5.02$ | 1, 39.14 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Chanel et al. (2012) [12] | 21 dyads | competitive and cooperative game play | RESP | Weighted coherence | Pooled | social empathy rating | Linear mixed model; C | $F = 5.33$ | 1, 57.23 |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | grounding & coordination | Regression modeling; U | $r = -0.15587$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | conflict | Regression modeling; U | $r = -0.145$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | convergence | Regression modeling; U | $r = 0.19341$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | confrontation & consensus building | Regression modeling; U | $r = -0.32673$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | coconstruction | Regression modeling; U | $r = -0.09532$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | emotion management | Regression modeling; U | $r = -0.17729$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | emotion modeling | Regression modeling; U | $r = 0.21546$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Pearson's correlation | Bipolar | transactivity | Regression modeling; U | $r = -0.34692$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | grounding & coordination | Regression modeling; U | $r = -0.26673$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | conflict | Regression modeling; U | $r = 0.003218$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | convergence | Regression modeling; U | $r = 0.005512$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | confrontation & consensus building | Regression modeling; U | $r = 0.37642$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | coconstruction | Regression modeling; U | $r = 0.086517$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | emotion management | Regression modeling; U | $r = -0.42769$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | emotion modeling | Regression modeling; U | $r = -0.1222$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; VLF | Pooled | transactivity | Regression modeling; U | $r = 0.16583$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | grounding & coordination | Regression modeling; U | $r = -0.085526$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | conflict | Regression modeling; U | $r = -0.28544$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | convergence | Regression modeling; U | $r = 0.40763$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | confrontation & consensus building | Regression modeling; U | $r = -0.13506$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | coconstruction | Regression modeling; U | $r = -0.073852$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | emotion management | Regression modeling; U | $r = -0.56629$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | emotion modeling | Regression modeling; U | $r = -0.02492$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; LF | Pooled | transactivity | Regression modeling; U | $r = -0.14745$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | grounding & coordination | Regression modeling; U | $r = -0.15398$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | conflict | Regression modeling; U | $r = -0.060808$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | convergence | Regression modeling; U | $r = 0.14445$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | confrontation & consensus building | Regression modeling; U | $r = -0.17911$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | coconstruction | Regression modeling; U | $r = -0.20463$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | emotion management | Regression modeling; U | $r = -0.32359$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | emotion modeling | Regression modeling; U | $r = -0.087582$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | R-R interval | Weighted coherence; HF | Pooled | transactivity | Regression modeling; U | $r = -0.23178$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | grounding & coordination | Regression modeling; U | $r = -0.35154$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | conflict | Regression modeling; U | $r = 0.28691$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | convergence | Regression modeling; U | $r = -0.38669$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | confrontation & consensus building | Regression modeling; U | $r = -0.029663$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | coconstruction | Regression modeling; U | $r = -0.05457$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | emotion management | Regression modeling; U | $r = 0.14432$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | emotion modeling | Regression modeling; U | $r = -0.098223$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | SCR | Pearson's correlation | Bipolar | transactivity | Regression modeling; U | $r = -0.071951$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | grounding & coordination | Regression modeling; U | $r = 0.11133$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | conflict | Regression modeling; U | $r = -0.030134$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | convergence | Regression modeling; U | $r = -0.049124$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | confrontation & consensus building | Regression modeling; U | $r = 0.018681$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | coconstruction | Regression modeling; U | $r = 0.13482$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | emotion management | Regression modeling; U | $r = 0.11144$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | emotion modeling | Regression modeling; U | $r = 0.062725$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; raw | Bipolar | transactivity | Regression modeling; U | $r = 0.15823$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | grounding & coordination | Regression modeling; U | $r = 0.27112$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | conflict | Regression modeling; U | $r = 0.0069463$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | convergence | Regression modeling; U | $r = 0.12108$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | confrontation & consensus building | Regression modeling; U | $r = -0.059747$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | coconstruction | Regression modeling; U | $r = -0.19946$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | emotion management | Regression modeling; U | $r = 0.065613$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | emotion modeling | Regression modeling; U | $r = -0.15522$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Pearson's correlation; filtered | Bipolar | transactivity | Regression modeling; U | $r = 0.16362$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | grounding & coordination | Regression modeling; U | $r = -0.042282$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | conflict | Regression modeling; U | $r = -0.24381$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | convergence | Regression modeling; U | $r = 0.22719$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | confrontation & consensus building | Regression modeling; U | $r = -0.15326$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | coconstruction | Regression modeling; U | $r = -0.077465$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | emotion management | Regression modeling; U | $r = -0.13974$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | emotion modeling | Regression modeling; U | $r = -0.10719$ | — |
| Chanel et al. (2013) [11] | 30 dyads | cooperative task | RESP | Weighted coherence | Pooled | transactivity | Regression modeling; U | $r = -0.13184$ | — |
| Chen et al. (2021) [13] | 137 dyads | shared positive emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = 0.096$ | 100 |
| Chen et al. (2021) [13] | 136 dyads | shared positive emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | affective valence at T1 | Pearson's correlation; U | $r = 0.078$ | 100 |
| Chen et al. (2021) [13] | 137 dyads | shared positive emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = 0.041$ | 100 |
| Chen et al. (2021) [13] | 131 dyads | shared negative emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | affective valence at T1 | Pearson's correlation; U | $r = 0.059$ | 100 |
| Chen et al. (2021) [13] | 132 dyads | shared negative emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = -0.013$ | 100 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Chen et al. (2021) [13] | 131 dyads | shared negative emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | affective valence at T1 | Pearson's correlation; U | $r = -0.17$ | 100 |
| Chen et al. (2021) [13] | 132 dyads | shared negative emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = -0.064$ | 100 |
| Chen et al. (2021) [13] | 135 dyads | shared neutral emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | affective valence at T1 | Pearson's correlation; U | $r = 0.041$ | 100 |
| Chen et al. (2021) [13] | 138 dyads | shared neutral emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = -0.097$ | 100 |
| Chen et al. (2021) [13] | 135 dyads | shared neutral emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | affective valence at T1 | Pearson's correlation; U | $r = -0.018$ | 100 |
| Chen et al. (2021) [13] | 138 dyads | shared neutral emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | Anti-phase | marital satisfaction at T1 | Pearson's correlation; U | $r = 0.039$ | 100 |
| Chen et al. (2021) [13] | 136 dyads | shared positive emotion during problem area discussion | SCL, R-R interval, BVP | Pearson's correlation | In-phase | affective valence at T1 | Pearson's correlation; U | $r = -0.14$ | 100 |
| Cheong et al. (2023) [14] | 27 dyads | watch TV show together | EDA | Pearson's correlation | Bipolar | sense of connection | Linear mixed-effects regression model; U | $r = 0.36$ | — |
| Chikersal et al. (2017) [15] | 52 dyads | collective intelligence task | SCR | Dynamic time warping | In-phase | group satisfaction | Pearson's correlation; U | $r = 0.325$ | — |
| Chikersal et al. (2017) [15] | 52 dyads | collective intelligence task | HR | Dynamic time warping | In-phase | group satisfaction | Pearson's correlation; U | $r = -0.002$ | — |
| Chikersal et al. (2017) [15] | 52 dyads | collective intelligence task | SCR | Dynamic time warping | In-phase | collective intelligence | Pearson's correlation; U | $r = -0.064$ | — |
| Chikersal et al. (2017) [15] | 52 dyads | collective intelligence task | HR | Dynamic time warping | In-phase | collective intelligence | Pearson's correlation; U | $r = -0.14$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | female's perceived individual empathy | Multiple linear regression; C | $t = -0.89$ | 21 |
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | male's perceived individual empathy | Multiple linear regression; C | $t = 0.95$ | 21 |
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | females' perceived dyadic empathy | Multiple linear regression; C | $t = -1.59$ | 21 |
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | male's perceived dyadic empathy | Multiple linear regression; C | $t = 1.33$ | 21 |
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | female's relationship satisfaction | Multiple linear regression; C | $t = 0.46$ | 21 |
| Coutinho et al. (2019) [16] | 32 couples | discussion of positive aspects | EDA | Cross-correlation | Pooled | male's relationship satisfaction | Multiple linear regression; C | $t = -0.55$ | 21 |
| Danyluck and Page-Gould (2019) [17] | 67 dyads | competitive and cooperative task with and without talking | CSI | Vector autoregression | Bipolar | perceived similarity | Multilevel modeling; U | $r = 0.195$ | — |
| Danyluck and Page-Gould (2019) [17] | 67 dyads | competitive and cooperative task with and without talking | CSI | Vector autoregression | Bipolar | friendship interest | Multilevel modeling; U | $r = 0.021$ | — |
| Danyluck and Page-Gould (2019) [17] | 67 dyads | competitive and cooperative task with and without talking | RSA | Vector autoregression | Bipolar | perceived similarity | Multilevel modeling; U | $r = -0.05$ | — |
| Danyluck and Page-Gould (2019) [17] | 67 dyads | competitive and cooperative task with and without talking | RSA | Vector autoregression | Bipolar | friendship interest | Multilevel modeling; U | $r = -0.041$ | — |
| Elkins et al. (2009) [23] | 10 teams of four | cooperative task | RSA | Pearson's correlation | Bipolar | team performance | t-test; U | $t = -2.31$ | 8 |
| Elkins et al. (2009) [23] | 10 teams of four | cooperative task | IBI | Directional agreement | In-phase | team performance | t-test; U | $t = 1.86$ | 4.62 |
| Elkins et al. (2009) [23] | 10 teams of four | cooperative task | RSA | Directional agreement | In-phase | team performance | t-test; U | $t = -1.68$ | 8 |
| Fogel-Yaakobi et al. (2023) [26] | 58 couples (women is in their third trimester) | asymmetric discussion of problem/concern | HR | Cross-correlation | In-phase | caregiver's supportive behaviors | Pearson's correlation + Multiple linear regressions; U | $r = -0.22$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Fogel-Yaakobi et al. (2023) [26] | 58 couples (women is in their third trimester) | asymmetric discussion of problem/concern | HR | Cross-correlation | In-phase | caregiver's attachment anxiety | Pearson's correlation + Multiple linear regressions; U | $r = -0.1$ | — |
| Fogel-Yaakobi et al. (2023) [26] | 58 couples (women is in their third trimester) | asymmetric discussion of problem/concern | HR | Cross-correlation | In-phase | caregiver's attachment avoidance | Pearson's correlation + Multiple linear regressions; U | $r = 0.3$ | — |
| Fusaroli et al. (2016) [27] | 30 participants | cooperative task | HR | CRQA; recurrence rate | Pooled | perceived relatedness | Mixed effects linear models; U | $r = -0.14$ | — |
| Fusaroli et al. (2016) [27] | 30 participants | cooperative task | HR | CRQA; average length of sequences repeated across time-series | Pooled | perceived relatedness | Mixed effects linear models; U | $r = 0.121$ | — |
| Fusaroli et al. (2016) [27] | 30 participants | cooperative task | HR | CRQA; recurrence rate | Pooled | perceived performance | Mixed effects linear models; U | $r = -0.41$ | — |
| Fusaroli et al. (2016) [27] | 30 participants | cooperative task | HR | CRQA; average length of sequences repeated across time-series | Pooled | perceived performance | Mixed effects linear models; U | $r = -0.17$ | — |
| Gibbs et al. (2023) [28] | 21 participants | play music together | HR | Pearson's correlation | Bipolar | awareness rating of flow state | Linear mixed models; C | $t = 2.14$ | 40.023 |
| Gibbs et al. (2023) [28] | 21 participants | play music together | HR | Pearson's correlation | Bipolar | absorption rating of flow state | Linear mixed models; C | $t = 2.073$ | 40.424 |
| Gibbs et al. (2023) [28] | 17 participants | play music together | EDA | Pearson's correlation | Bipolar | awareness rating of flow state | Linear mixed models; C | $t = 2.605$ | 32.028 |
| Gordon et al. (2020) [30] | 47 triads | drumming task | IBI | Cross-correlation | In-phase | group performance | Hierarchical linear regression modeling; U | $r = 0.306$ | — |
| Gordon et al. (2020) [30] | 47 triads | drumming task | IBI | Cross-correlation | In-phase | perceived group cohesion | Hierarchical linear regression modeling; U | $r = 0.236$ | — |
| Greenberg et al. (2022) [31] | 48 triads | drumming task | EDA | Cross-correlation | Bipolar | group efficacy | Linear mixed model analysis; C | $t = -0.8$ | 42.76 |
| Grochowska et al. (2024) [32] | 15 dyads | free interaction | HRV | CRQA; Max Recurrence | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.502$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | free interaction | HRV | CRQA; %Rec | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.254$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | free interaction | HRV | CRQA; %DET | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.143$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Grochowska et al. (2024) [32] | 15 dyads | free interaction | HRV | CRQA; Maximal length | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.11$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | soothing after stress | HRV | CRQA; Max Recurrence | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = 0.009$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | soothing after stress | HRV | CRQA; %Rec | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = 0.225$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | soothing after stress | HRV | CRQA; %DET | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.102$ | — |
| Grochowska et al. (2024) [32] | 15 dyads | soothing after stress | HRV | CRQA; Maximal length | Bipolar | affective touch | Spearman's rho correlation coefficient; U | $\rho = -0.082$ | — |
| Han et al. (2019) [33] | 150 dyads | collaborative drawing | IBI | Cross-correlation (lag = 0) | Bipolar | psychological unavailability | Pearson's correlation; U | $r = -0.18$ | — |
| Han et al. (2019) [33] | 150 dyads | collaborative drawing | IBI | Cross-correlation (lag = 0) | Bipolar | psychological control | Pearson's correlation; U | $r = -0.12$ | — |
| Han et al. (2019) [33] | 150 dyads | conflict discussion | IBI | Cross-correlation (lag = 0) | Bipolar | psychological unavailability | Pearson's correlation; U | $r = -0.03$ | — |
| Han et al. (2019) [33] | 150 dyads | conflict discussion | IBI | Cross-correlation (lag = 0) | Bipolar | psychological control | Pearson's correlation; U | $r = -0.11$ | — |
| Han et al. (2019) [33] | 150 dyads | collaborative drawing | IBI | Cross-correlation (lag = 0) | Bipolar | psychological unavailability | Multivariate linear regression; C | $t = -2.02$ | 128 |
| Han et al. (2019) [33] | 150 dyads | collaborative drawing | IBI | Cross-correlation (lag = 0) | Bipolar | psychological control | Multivariate linear regression; C | $t = -1.32$ | 128 |
| Han et al. (2019) [33] | 150 dyads | conflict discussion | IBI | Cross-correlation (lag = 0) | Bipolar | psychological unavailability | Multivariate linear regression; C | $t = 0.94$ | 124 |
| Han et al. (2019) [33] | 150 dyads | conflict discussion | IBI | Cross-correlation (lag = 0) | Bipolar | psychological control | Multivariate linear regression; C | $t = -2.09$ | 126 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | R-R interval | Cross-correlation (lag = 0) | Bipolar | task completion time | Multiple regression; C | $t = -4.98$ | 60 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | EDA | Weighted coherence | Pooled | task completion time | Multiple regression; C | $t = -2.29$ | 60 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | R-R interval | Weighted coherence | Pooled | task completion time | Multiple regression; C | $t = -2.81$ | 60 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | R-R interval | Cross-correlation (lag = 0) | Bipolar | task error rate | Multiple regression; C | $t = -3.98$ | 60 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | EDA | Weighted coherence | Pooled | task error rate | Multiple regression; C | $t = -2.06$ | 60 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Henning et al. (2001) [34] | 16 dyads | cooperative task | R-R interval | Weighted coherence | Pooled | task error rate | Multiple regression; C | $t = -2.61$ | 60 |
| Henning et al. (2001) [34] | 16 dyads | cooperative task | RESP | Weighted coherence | Pooled | task error rate | Multiple regression; C | $t = -2.17$ | 60 |
| Jospe et al. (2020) [36] | 55 dyads | listening of emotional story | HR | Cross-correlation | Bipolar | self-reported empathy (EQ) | Pearson's correlation; U | $r = 0.042$ | — |
| Jospe et al. (2020) [36] | 55 dyads | listening of emotional story | HR | Cross-correlation | Bipolar | self-reported empathy (IRI) | Pearson's correlation; U | $r = -0.212$ | — |
| Jospe et al. (2020) [36] | 55 dyads | listening of emotional story | HR | Cross-correlation | Bipolar | empathic accuracy | Pearson's correlation; U | $r = 0.245$ | — |
| Jospe et al. (2020) [36] | 55 dyads | listening of emotional story | HR | Cross-correlation | Bipolar | empathic accuracy for specific emotions | Pearson's correlation; U | $r = 0.268$ | — |
| Järvelä et al,. (2014) [37] | 41 dyads | cooperative and competitive game | HRV | Cross-correlation (lag = 0) | Bipolar | mutual understanding | Linear Mixed model; C | $F = 8.589$ | 1, — |
| Järvelä et al,. (2014) [37] | 41 dyads | cooperative and competitive game | R-R interval | Weighted coherence | Pooled | behavioral involvement | Linear Mixed model; C | $F = 9.025$ | 1, — |
| Järvelä et al,. (2014) [37] | 41 dyads | cooperative and competitive game | HRV | Cross-correlation (lag = 0) | Bipolar | behavioral involvement | Linear Mixed model; C | $F = 5.119$ | 1, — |
| Järvelä et al,. (2014) [37] | 41 dyads | cooperative and competitive game | R-R interval | Cross-correlation (lag = 0) | Bipolar | emptional empathy | Linear Mixed model; C | $F = 8.641$ | 1, — |
| Levenson and Gottman (1983) [41] | 30 couples | event-of-the-day discussion | IBI, PTT, SCL | Vector autoregression | Pooled | marital satisfaction | Pearson's correlation Hierarchical Multiple Regression Analysis; U | $r = -0.04$ | — |
| Levenson and Gottman (1983) [41] | 30 couples | problem-area discussion | IBI, PTT, SCL | Vector autoregression | Pooled | marital satisfaction | Pearson's correlation Hierarchical Multiple Regression Analysis; U | $r = -0.31$ | — |
| Levenson and Ruef (1992) [42] | 31 participants | rate positive affect while watching event-of-the-day conversation | IBI, PTT, SCL | Vector autoregression | Bipolar | empathic accuracy | Pearson's correlation; U | $r = 0.16$ | — |
| Levenson and Ruef (1992) [42] | 31 participants | rate negative affect while watching event-of-the-day conversation | IBI, PTT, SCL | Vector autoregression | Bipolar | empathic accuracy | Pearson's correlation; U | $r = 0.46$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Levenson and Ruef (1992) [42] | 31 participants | rate positive affect while watching problem area conversation | IBI, PTT, SCL | Vector autoregression | Bipolar | empathic accuracy | Pearson's correlation; U | $r = 0.23$ | — |
| Levenson and Ruef (1992) [42] | 31 participants | rate negative affect while watching problem-area conversation | IBI, PTT, SCL | Vector autoregression | Bipolar | empathic accuracy | Pearson's correlation; U | $r = 0.54$ | — |
| Liu et al. (2018) [43] | 48 couples | free play with their children | RSA | Partial Directed Coherence | Pooled | wife's reported conflict [Target: husband] | Actor-partner interdependence model; C | $r_{\text{partial}} = 0.33$ | — |
| Liu et al. (2018) [43] | 48 couples | free play with their children | RSA | Partial Directed Coherence | Pooled | husband's reported conflict [Target: husband] | Actor-partner interdependence model; C | $r_{\text{partial}} = 0.13$ | — |
| Liu et al. (2018) [43] | 48 couples | free play with their children | RSA | Partial Directed Coherence | Pooled | wife's reported conflict [Target: wife] | Actor-partner interdependence model; C | $r_{\text{partial}} = -0.05$ | — |
| Liu et al. (2018) [43] | 48 couples | free play with their children | RSA | Partial Directed Coherence | Pooled | husband's reported conflict [Target: wife] | Actor-partner interdependence model; C | $r_{\text{partial}} = -0.01$ | — |
| Liu et al. (2021) [44] | 5 groups | collaborative problem solving | EDA | Single session index | In-phase | collaboration level | Group comparison between collaborative and non-collaborative groups; U | $t = 2.781$ | 27 |
| Marci and Orr (2006) [46] | 20 dyads | therapist-client conversation | EDA | Single session index | In-phase | emotional distance | t-test of group comparison; U | $t = 3.41$ | 18 |
| Marci et al. (2007) [45] | 20 dyads | therapy session | EDA | Single session index | In-phase | perceived empathy | Pearson's correlation; U | $r = 0.47$ | — |
| Messina et al. (2013) [50] | 39 dyads | simulated listener-patient conversation | EDA | Cross-correlation | Bipolar | perceived empathy by listener | Pearson's correlation; U | $r = 0.37$ | — |
| Montague et al. (2014) [51] | 24 dyads | collaborative task | EDA | Directional agreement | In-phase | task performance | Linear mixed effects model; C | $t = 2.541$ | 52 |
| Montague et al. (2014) [51] | 24 dyads | collaborative task | EDA | Signal matching | In-phase | task performance | Linear mixed effects model; C | $t = -2.16$ | 52 |
| Noy et al. (2015) [53] | 7 dyads | cooperative task | HR | Pearson's correlation | Bipolar | subjective togetherness | Pearson's correlation; U | $r = 0.38$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent positivity toward child | Pearson's correlation; U | $r = 0.01$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent negativity toward child | Pearson's correlation; U | $r = 0.13$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent respect for autonomy | Pearson's correlation; U | $r = 0.07$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parental engagement | Pearson's correlation; U | $r = 0.01$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent intrusiveness | Pearson's correlation; U | $r = 0.09$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent use of reasoning | Pearson's correlation; U | $r = 0.15$ | — |
| Oshri et al. (2023) [54] | 101 dyads | discussion of topic of disagreement | RSA | Structural equation modeling | Bipolar | parent coercion | Pearson's correlation; U | $r = -0.02$ | — |
| Prochazkova et al. (2022) [56] | 54 dyads | free inteeraction | HR | Cross-correlation | Bipolar | attraction | Multilevel modeling; C | $t = 2.34$ | 324 |
| Prochazkova et al. (2022) [56] | 54 dyads | free inteeraction | EDA | Cross-correlation | Bipolar | attraction | Multilevel modeling; C | $t = 2.67$ | 324 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | RSA | Multilevel modeling | Bipolar | empathic accuracy | Multilevel modeling; C | $t = 0.1$ | 127.6 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | IBI | Multilevel modeling | Bipolar | empathic accuracy | Multilevel modeling; C | $t = -0.5$ | 135.97 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | SCL | Multilevel modeling | Bipolar | empathic accuracy | Multilevel modeling; C | $t = -1.16$ | 134.6 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | RSA | Multilevel modeling | Bipolar | emotion contagion | Multilevel modeling; C | $t = -0.75$ | 134.27 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | IBI | Multilevel modeling | Bipolar | emotion contagion | Multilevel modeling; C | $t = 0.75$ | 129.65 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | SCL | Multilevel modeling | Bipolar | emotion contagion | Multilevel modeling; C | $t = -0.53$ | 141.37 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | RSA | Multilevel modeling | Bipolar | compassion | Multilevel modeling; C | $t = 1$ | 139.17 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | IBI | Multilevel modeling | Bipolar | compassion | Multilevel modeling; C | $t = -1.39$ | 134.7 |
| Quaiser et al. (2023) [57] | 95 couples | disclosure of negative experience | SCL | Multilevel modeling | Bipolar | compassion | Multilevel modeling; C | $t = -0.34$ | 146.99 |
| Robinson et al. (1982) [59] | 21 dyads | councelling | SCR | Delayed coincidence count; small but rapid response | In-phase | perceived empathy | Pearson's correlation; U | $r = 0.453$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Robinson et al. (1982) [59] | 21 dyads | councelling | SCR | Delayed coincidence count; large amplitude long latency | In-phase | perceived empathy | Pearson's correlation; U | $r = 0.033$ | — |
| Robinson et al. (1982) [59] | 21 dyads | councelling | SCR | Delayed coincidence count; large amplitude short latency | In-phase | perceived empathy | Pearson's correlation; U | $r = 0.617$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | negative affect [Target: proband] | Pearson's correlation; U | $r = -0.24$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | negative affect [Target: partner] | Pearson's correlation; U | $r = -0.03$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | perception of warmth [Target: proband] | Pearson's correlation; U | $r = 0.2$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | perception of warmth [Target: partner] | Pearson's correlation; U | $r = 0.04$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | relationship satisfaction (baseline) [Target: proband] | Pearson's correlation; U | $r = 0.05$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | relationship satisfaction (baseline) [Target: partner] | Pearson's correlation; U | $r = 0.21$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | relationship satisfaction (12-mo follow-up) [Target: proband] | Pearson's correlation; U | $r = -0.11$ | — |
| Schreiber et al. (2020) [62] | 99 couples | discussion on topic of disagreement | R-R interval | Differential equations | Bipolar | relationship satisfaction (12-mo follow-up) [Target: partner] | Pearson's correlation; U | $r = 0.17$ | — |
| Shimshock et al. (2024) [63] | 79 couples | one partner tells pleasant news and the other listens | R-R interval | Multilevel modeling | Bipolar | neglect/withdrawal [Target: discloser] | Mixed effects model; C | $t = 1.06$ | 60.3 |
| Shimshock et al. (2024) [63] | 79 couples | one partner tells pleasant news and the other listens | R-R interval | Multilevel modeling | Bipolar | neglect/withdrawal [Target: listener] | Mixed effects model; C | $t = -1.94$ | 59.6 |
| Shimshock et al. (2024) [63] | 79 couples | one partner tells pleasant news and the other listens | R-R interval | Multilevel modeling | Bipolar | behavioral positivity [Target: discloser] | Mixed effects model; C | $t = 1.93$ | 56.7 |
| Shimshock et al. (2024) [63] | 79 couples | one partner tells pleasant news and the other listens | R-R interval | Multilevel modeling | Bipolar | behavioral positivity [Target: listener] | Mixed effects model; C | $t = 2.14$ | 56.7 |
| Snijdewint and Scheepers (2023) [64] | 37 groups | cooperative game play | HR | Dynamic time warping | In-phase | task performance | Pearson's correlation; U | $r = -0.27$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Snijdewint and Scheepers (2023) [64] | 37 groups | cooperative game play | HR | Dynamic time warping | In-phase | flow state | Pearson's correlation; U | $r = 0.07$ | — |
| Snijdewint and Scheepers (2023) [64] | 37 groups | cooperative game play | HR | Dynamic time warping | In-phase | perceived group cohesion | Pearson's correlation; U | $r = 0$ | — |
| Snijdewint and Scheepers (2023) [64] | 34 groups | cooperative game play | PEP | Dynamic time warping | In-phase | task performance | Pearson's correlation; U | $r = -0.47$ | — |
| Snijdewint and Scheepers (2023) [64] | 34 groups | cooperative game play | PEP | Dynamic time warping | In-phase | flow state | Pearson's correlation; U | $r = -0.32$ | — |
| Snijdewint and Scheepers (2023) [64] | 34 groups | cooperative game play | PEP | Dynamic time warping | In-phase | perceived group cohesion | Pearson's correlation; U | $r = -0.22$ | — |
| Snijdewint and Scheepers (2023) [64] | 33 groups | cooperative game play | CO | Dynamic time warping | In-phase | task performance | Pearson's correlation; U | $r = -0.35$ | — |
| Snijdewint and Scheepers (2023) [64] | 33 groups | cooperative game play | CO | Dynamic time warping | In-phase | flow state | Pearson's correlation; U | $r = -0.17$ | — |
| Snijdewint and Scheepers (2023) [64] | 33 groups | cooperative game play | CO | Dynamic time warping | In-phase | perceived group cohesion | Pearson's correlation; U | $r = -0.04$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | task performance | Pearson's correlation; U | $r = 0.36$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | game experience | Pearson's correlation; U | $r = 0.3$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | perceived group cohesion | Pearson's correlation; U | $r = 0.19$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | perceived trust | Pearson's correlation; U | $r = -0.01$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | perceived collective efficacy | Pearson's correlation; U | $r = 0.08$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Cross-correlation (lag = 0) | Bipolar | perceived communication effectiveness | Pearson's correlation; U | $r = -0.06$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | task performance | Pearson's correlation; U | $r = -0.3$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | game experience | Pearson's correlation; U | $r = -0.25$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | perceived group cohesion | Pearson's correlation; U | $r = -0.29$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | perceived trust | Pearson's correlation; U | $r = -0.32$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | perceived collective efficacy | Pearson's correlation; U | $r = -0.21$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | CRQA | Pooled | perceived communication effectiveness | Pearson's correlation; U | $r = -0.29$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | task performance | Pearson's correlation; U | $r = -0.47$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | game experience | Pearson's correlation; U | $r = -0.27$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | perceived group cohesion | Pearson's correlation; U | $r = -0.44$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | perceived trust | Pearson's correlation; U | $r = -0.56$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | perceived collective efficacy | Pearson's correlation; U | $r = -0.28$ | — |
| Strang et al. (2014) [65] | 40 dyads | cooperative game play | IBI | Inverse cross-fuzzy entropy | In-phase | perceived communication effectiveness | Pearson's correlation; U | $r = -0.46$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (client) [Target: female client + male therapist] | Pearson's correlation; U | $r = 0.14$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (client) [Target: female client + female therapist] | Pearson's correlation; U | $r = 0.02$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (client) [Target: male client + male therapist] | Pearson's correlation; U | $r = -0.44$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (client) [Target: male client + female therapist] | Pearson's correlation; U | $r = 0.16$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (therapist) [Target: female client + male therapist] | Pearson's correlation; U | $r = 0.18$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (therapist) [Target: female client + female therapist] | Pearson's correlation; U | $r = 0.29$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (therapist) [Target: male client + male therapist] | Pearson's correlation; U | $r = -0.77$ | — |
| Tourunen et al. (2020) [67] | 12 groups | couple therapy | EDA | Cross-correlation | In-phase | therapeutic alliance (therapist) [Target: male client + female therapist] | Pearson's correlation; U | $r = -0.11$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Pooled | client's perceived alliance | Multivariate regression models; C | $t = -0.16$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Pooled | therapist's perceived alliance | Multivariate regression models; C | $t = -0.08$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Pooled | therapist's perceived cooperation | Multivariate regression models; C | $t = -1.18$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Bipolar | client's perceived alliance | Mixed effects linear models; C | $t = 2.87$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Bipolar | therapist's perceived alliance | Mixed effects linear models; C | $t = 1.05$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Cross-correlation | Bipolar | therapist's perceived cooperation | Mixed effects linear models; C | $t = -1.84$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Single session index | In-phase | client's perceived alliance | Multivariate regression models; C | $t = -0.42$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Single session index | In-phase | therapist's perceived alliance | Multivariate regression models; C | $t = -0.45$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | RESP | Single session index | In-phase | therapist's perceived cooperation | Multivariate regression models; C | $t = 0.46$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Pooled | client's perceived alliance | Multivariate regression models; C | $t = 0.01$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Pooled | therapist's perceived alliance | Multivariate regression models; C | $t = 0.76$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Pooled | therapist's perceived cooperation | Multivariate regression models; C | $t = -0.13$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Bipolar | client's perceived alliance | Multivariate regression models; C | $t = -0.8$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Bipolar | therapist's perceived alliance | Multivariate regression models; C | $t = -0.56$ | 38 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Cross-correlation | Bipolar | therapist's perceived cooperation | Multivariate regression models; C | $t = -2$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | client's perceived alliance | Multivariate regression models; C | $t = -0.41$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | therapist's perceived alliance | Multivariate regression models; C | $t = -0.84$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | therapist's perceived cooperation | Multivariate regression models; C | $t = -0.96$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | client's perceived alliance | Mixed effects linear models; C | $t = -1.21$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | therapist's perceived alliance | Mixed effects linear models; C | $t = 0.44$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HR | Single session index | In-phase | therapist's perceived cooperation | Mixed effects linear models; C | $t = -0.96$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Pooled | client's perceived alliance | Multivariate regression models; C | $t = -0.55$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Pooled | therapist's perceived alliance | Multivariate regression models; C | $t = 0.61$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Pooled | therapist's perceived cooperation | Multivariate regression models; C | $t = -1.43$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Bipolar | client's perceived alliance | Multivariate regression models; C | $t = -0.62$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Bipolar | therapist's perceived alliance | Multivariate regression models; C | $t = 2.41$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Cross-correlation | Bipolar | therapist's perceived cooperation | Multivariate regression models; C | $t = -0.92$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | client's perceived alliance | Multivariate regression models; C | $t = 1.2$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | therapist's perceived alliance | Multivariate regression models; C | $t = -0.36$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | therapist's perceived cooperation | Multivariate regression models; C | $t = -0.18$ | 38 |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | client's perceived alliance | Mixed effects linear models; C | $t = 0.8$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | therapist's perceived alliance | Mixed effects linear models; C | $t = 0.32$ | — |
| Tschacher and Meier (2020) [68] | 47 dyads | therapy session | HRV | Single session index | In-phase | therapist's perceived cooperation | Mixed effects linear models; C | $t = -2.13$ | — |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived bond | Hierarchical regression models; C | $t = -0.19$ | 198.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived agreement on goal | Hierarchical regression models; C | $t = 1.42$ | 253.9 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived agreement on task | Hierarchical regression models; C | $t = -0.37$ | 262 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived bond | Hierarchical regression models; C | $t = 1.08$ | 278 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived agreement on goal | Hierarchical regression models; C | $t = -0.62$ | 278.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived agreement on task | Hierarchical regression models; C | $t = 0.76$ | 276.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived bond | Hierarchical regression models; C | $t = -0.77$ | 95.13 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived agreement on goal | Hierarchical regression models; C | $t = -0.48$ | 146.6 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived agreement on task | Hierarchical regression models; C | $t = 1.92$ | 180 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived bond | Hierarchical regression models; C | $t = 0.31$ | 247.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived agreement on goal | Hierarchical regression models; C | $t = -0.03$ | 264.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived agreement on task | Hierarchical regression models; C | $t = 0.01$ | 272.4 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived bond [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.09$ | 166.1 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived agreement on goal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.13$ | 230.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived agreement on task [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.87$ | 245.2 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived bond [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 2.08$ | 274.2 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived agreement on goal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.63$ | 279 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived agreement on task [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.07$ | 278.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived bond [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.93$ | 156.6 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived agreement on goal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.94$ | 228.1 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived agreement on task [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -1.02$ | 243.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived bond [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 2.08$ | 273.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived agreement on goal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -1.92$ | 278.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived agreement on task [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 1.81$ | 278.7 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived depth | Hierarchical regression models; C | $t = 1.37$ | 96.68 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived smoothness | Hierarchical regression models; C | $t = -0.76$ | 139.9 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's post session positivity | Hierarchical regression models; C | $t = 1.24$ | 137.9 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's post session arousal | Hierarchical regression models; C | $t = 0.14$ | 135 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived depth | Hierarchical regression models; C | $t = 0.36$ | 229.6 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived smoothness | Hierarchical regression models; C | $t = 1.14$ | 196.2 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's post session positivity | Hierarchical regression models; C | $t = -0.15$ | 201.6 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | thrapist's post session arousal | Hierarchical regression models; C | $t = -2.28$ | 132.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived depth | Hierarchical regression models; C | $t = 1.98$ | 227.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived smoothness | Hierarchical regression models; C | $t = 0.28$ | 259.4 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's post session positivity | Hierarchical regression models; C | $t = -0.84$ | 255.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's post session arousal | Hierarchical regression models; C | $t = -0.98$ | 257.9 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived depth | Hierarchical regression models; C | $t = 0.53$ | 266 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived smoothness | Hierarchical regression models; C | $t = -0.83$ | 265.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's post session positivity | Hierarchical regression models; C | $t = 2.01$ | 264.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | thrapist's post session arousal | Hierarchical regression models; C | $t = -0.58$ | 246.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived depth [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.36$ | 201.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's perceived smoothness [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 1.06$ | 245.7 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's post session positivity [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.25$ | 240.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | client's post session arousal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.25$ | 243.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived depth [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 1.09$ | 264.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's perceived smoothness [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.47$ | 264.5 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | therapist's post session positivity [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -1.5$ | 259.3 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Pooled | thrapist's post session arousal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.57$ | 226.8 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived depth [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.73$ | 156.6 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's perceived smoothness [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 1.33$ | 214 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's post session positivity [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.66$ | 207.1 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | client's post session arousal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.07$ | 210.4 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived depth [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -1.06$ | 257.7 |

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Table 1 (continued)

| Study | Sample | Condition | Measure | Linkage method | Linkage coding | Outcome / target | Association analysis | Statistic | *df* |
|---|---|---|---|---|---|---|---|---|---|
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's perceived smoothness [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.64$ | 254 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | therapist's post session positivity [Target: client's leading synchrony] | Hierarchical regression models; C | $t = -0.36$ | 244.9 |
| Tschacher et al. (2025) [69] | 21 dyads | therapy session | EDA | Cross-correlation | Bipolar | thrapist's post session arousal [Target: client's leading synchrony] | Hierarchical regression models; C | $t = 0.55$ | 189.8 |
| West et al. (2020) [71] | 87 dyads | conflict discussion | RSA | Multilevel modeling | Bipolar | dyadic positivity | Pearson's correlation; U | $r = -0.09$ | — |
| West et al. (2020) [71] | 87 dyads | conflict discussion | RSA | Multilevel modeling | Bipolar | dyadic negativity | Pearson's correlation; U | $r = 0.01$ | — |
| West et al. (2020) [71] | 87 dyads | conflict discussion | RSA | Multilevel modeling | Bipolar | dyadic engagement | Pearson's correlation; U | $r = 0.07$ | — |
| Wilson et al. (2018) [72] | 43 couples | conflict discussion | HRV | Cross-correlation | Bipolar | negative affect | Pearson's correlation; U | $\rho = 0.14$ | — |
| Wilson et al. (2018) [72] | 43 couples | conflict discussion | HRV | Cross-correlation | Bipolar | momentary closeness | Pearson's correlation; U | $\rho = -0.46$ | — |
| Wilson et al. (2018) [72] | 43 couples | conflict discussion | HRV | Cross-correlation | Bipolar | negative interaction behaviors | Pearson's correlation; U | $\rho = 0.11$ | — |
| Wilson et al. (2018) [72] | 43 couples | conflict discussion | HRV | Cross-correlation | Bipolar | marital satisfaction | Pearson's correlation; U | $\rho = -0.05$ | — |
| Wilson et al. (2018) [72] | 43 couples | conflict discussion | HRV | Cross-correlation | Bipolar | general closeness | Pearson's correlation; U | $\rho = -0.06$ | — |
| Woltering et al. (2015) [74] | 99 dyads | positive and negative discussion | R-R interval | Structural heteroscedastic measurement-error model | Pooled | dyadic attunement | ANOVA and ANCOVA; C | $F = 4.18$ | —, 95 |
| Woltering et al. (2015) [74] | 96 dyads | positive and negative discussion | R-R interval | Structural heteroscedastic measurement-error model | Pooled | level of repair | ANOVA and ANCOVA; C | $F = 3.45$ | —, 91 |
| Xu et al. (2024) [75] | 123 dyads | conflict discussion | RSA | Multilevel modeling | Bipolar | relationship quality | Pearson's correlation; U | $r = 0.04$ | — |
| Xu et al. (2024) [75] | 123 dyads | conflict discussion | RSA | Multilevel modeling | Bipolar | parental emotional support | Pearson's correlation; U | $r = 0.08$ | — |
| Zerwas et al. (2021) [76] | 48 dyads | listening recorded speech | R-R interval | Cross-correlation | Bipolar | empathic accuracy | Multilevel modeling; C | $t = -0.35$ | 44 |
| Zerwas et al. (2021) [76] | 48 dyads | listening recorded speech | Finger Pulse Amplitude | Cross-correlation | Bipolar | empathic accuracy | Multilevel modeling; C | $t = -0.57$ | 42 |
| Zerwas et al. (2021) [76] | 48 dyads | listening recorded speech | SCL | Cross-correlation | Bipolar | empathic accuracy | Multilevel modeling; C | $t = -0.9$ | 38 |

*Notes.* C and U reproduce `Corrected = TRUE` and `Corrected = FALSE`, respectively.

Table 2: Moderator analyses for unadjusted effect sizes.

| Moderator | Category | $n$ | $k$ | Estimated $r$ | 95% CI | $p_{\text{mod}}$ |
|---|---|---|---|---|---|---|
| ANS branch | No distinction | 19 | 147 | 0.056 | $[-0.019, 0.130]$ | 0.221 |
| | PNS | 8 | 31 | $-0.063$ | $[-0.178, 0.055]$ | |
| | SNS | 14 | 36 | 0.070 | $[-0.108, 0.243]$ | |
| Stressful context | No | 24 | 157 | 0.043 | $[-0.050, 0.135]$ | 0.618 |
| | Yes | 12 | 57 | 0.016 | $[-0.051, 0.083]$ | |
| Helping context | No | 28 | 194 | 0.010 | $[-0.042, 0.063]$ | 0.097 |
| | Yes | 7 | 20 | 0.227 | $[-0.055, 0.475]$ | |
| Goal-oriented context | No | 21 | 79 | 0.057 | $[-0.017, 0.131]$ | 0.177 |
| | Yes | 14 | 135 | $-0.017$ | $[-0.108, 0.074]$ | |
| Relationship type | Couples | 5 | 30 | $-0.027$ | $[-0.200, 0.147]$ | 0.101 |
| | Parent-child | 7 | 29 | 0.041 | $[-0.054, 0.135]$ | |
| | Strangers | 6 | 16 | 0.100 | $[-0.081, 0.275]$ | |
| | Teammates | 11 | 125 | $-0.044$ | $[-0.168, 0.082]$ | |
| | Therapist-client | 5 | 14 | 0.365 | $[0.060, 0.608]$ | |
| Affective context | Mixed | 6 | 18 | 0.259 | $[-0.080, 0.544]$ | 0.680 |
| | Negative | 12 | 43 | 0.014 | $[-0.069, 0.098]$ | |
| | Neutral | 4 | 73 | $-0.029$ | $[-0.164, 0.108]$ | |
| | Not explicitly valenced | 16 | 74 | 0.005 | $[-0.104, 0.115]$ | |
| | Positive | 2 | 6 | 0.006 | $[-0.379, 0.389]$ | |
| Interaction format | Clinical or helping | 7 | 20 | 0.244 | $[-0.080, 0.521]$ | 0.564 |
| | Goal oriented joint activity | 14 | 135 | $-0.018$ | $[-0.108, 0.072]$ | |
| | Shared stimulus / passive observation | 3 | 9 | 0.222 | $[-0.412, 0.712]$ | |
| | Structured discussion | 9 | 44 | $-0.008$ | $[-0.075, 0.059]$ | |
| | Unstructured interaction | 3 | 6 | 0.100 | $[-0.950, 0.967]$ | |
| Outcome category | Affect | 7 | 17 | 0.038 | $[-0.120, 0.194]$ | 0.194 |
| | Clinical alliance | 1† | 8 | $-0.100$ | $[-0.100, -0.100]$ | |
| | Collective performance | 8 | 33 | $-0.086$ | $[-0.251, 0.085]$ | |
| | Empathy understanding | 6 | 15 | 0.205 | $[-0.065, 0.446]$ | |
| | Individual relational characteristics | 1† | 2 | $-0.046$ | $[-0.585, 0.521]$ | |
| | Interaction behavior | 9 | 88 | 0.021 | $[-0.068, 0.110]$ | |
| | Relationship quality | 14 | 51 | 0.015 | $[-0.071, 0.101]$ | |
| Linkage direction | Anti-phase | 1† | 6 | $-0.031$ | $[-0.263, 0.206]$ | 0.553 |
| | Bipolar | 21 | 102 | 0.076 | $[-0.007, 0.157]$ | |
| | In-phase | 14 | 62 | $-0.040$ | $[-0.226, 0.149]$ | |
| | Pooled together | 4 | 44 | $-0.013$ | $[-0.246, 0.222]$ | |
| Lag considered | No | 17 | 137 | 0.054 | $[-0.007, 0.115]$ | 0.262 |
| | Yes | 19 | 77 | $-0.001$ | $[-0.092, 0.090]$ | |
| Temporal dependence adjusted | No | 28 | 192 | 0.019 | $[-0.043, 0.082]$ | 0.437 |
| | Yes | 6 | 22 | 0.066 | $[-0.070, 0.200]$ | |
| Linkage method family | Coherence based | 1† | 32 | 0.059 | $[-0.037, 0.154]$ | 0.233 |
| | Correlation based | 21 | 96 | 0.089 | $[-0.008, 0.185]$ | |
| | Local concordance | 2 | 5 | 0.436 | $[-0.277, 0.839]$ | |
| | Model based coupling | 8 | 32 | 0.042 | $[-0.045, 0.129]$ | |
| | Pattern similarity | 6 | 49 | $-0.238$ | $[-0.491, 0.052]$ | |

$n$: distinct studies; $k$: effect sizes. Estimates and 95% CIs are back-transformed from Fisher's $z$ using single-moderator multilevel models with study-clustered CR2 inference. $p_{\text{mod}}$ tests differences across categories, not whether individual means differ from zero.

Table 3: Moderator analyses for adjusted effect sizes.

| Moderator | Category | $n$ | $k$ | Estimated $r$ | 95% CI | $p_{\text{mod}}$ |
|---|---|---|---|---|---|---|
| ANS branch | No distinction | 11 | 64 | 0.068 | $[-0.033, 0.169]$ | 0.750 |
| | PNS | 2 | 4 | 0.095 | $[-0.109, 0.292]$ | |
| | SNS | 10 | 74 | 0.060 | $[-0.040, 0.158]$ | |
| Stressful context | No | 12 | 51 | 0.097 | $[-0.005, 0.198]$ | 0.062 |
| | Yes | 4 | 91 | 0.001 | $[-0.059, 0.061]$ | |
| Helping context | No | 12 | 53 | 0.095 | $[-0.014, 0.201]$ | 0.136 |
| | Yes | 3 | 89 | $-0.007$ | $[-0.110, 0.096]$ | |
| Goal-oriented context | No | 8 | 106 | 0.030 | $[-0.039, 0.098]$ | 0.191 |
| | Yes | 8 | 36 | 0.114 | $[-0.044, 0.265]$ | |
| Relationship type | Couples | 3 | 19 | 0.035 | $[-0.292, 0.355]$ | 0.996 |
| | Friends | 1† | 3 | $-0.090$ | $[-0.392, 0.229]$ | |
| | Parent-child | 1† | 4 | 0.100 | $[-0.048, 0.244]$ | |
| | Strangers | 2 | 5 | 0.037 | $[-0.818, 0.841]$ | |
| | Teammates | 6 | 31 | 0.169 | $[-0.069, 0.389]$ | |
| | Therapist-client | 2 | 80 | 0.002 | $[-0.316, 0.319]$ | |
| Affective context | Mixed | 2 | 80 | 0.009 | $[-0.219, 0.237]$ | 0.826 |
| | Negative | 2 | 11 | 0.011 | $[-0.387, 0.404]$ | |
| | Neutral | 1† | 3 | $-0.049$ | $[-0.120, 0.022]$ | |
| | Not explicitly valenced | 9 | 38 | 0.136 | $[0.020, 0.249]$ | |
| | Positive | 2 | 10 | 0.103 | $[-0.713, 0.800]$ | |
| Interaction format | Clinical or helping | 3 | 89 | $-0.009$ | $[-0.107, 0.089]$ | 0.273 |
| | Goal oriented joint activity | 8 | 36 | 0.121 | $[-0.043, 0.278]$ | |
| | Shared stimulus / passive observation | 1† | 3 | $-0.090$ | $[-0.397, 0.234]$ | |
| | Structured discussion | 3 | 12 | 0.053 | $[-0.170, 0.271]$ | |
| | Unstructured interaction | 1† | 2 | 0.138 | $[0.138, 0.138]$ | |
| Outcome category | Affect | 3 | 21 | 0.020 | $[-0.041, 0.080]$ | – |
| | Clinical alliance | 2 | 48 | 0.051 | $[-0.007, 0.109]$ | |
| | Collective performance | 4 | 27 | 0.068 | $[-0.412, 0.519]$ | |
| | Empathy understanding | 4 | 14 | 0.040 | $[-0.150, 0.228]$ | |
| | Interaction behavior | 3 | 24 | 0.078 | $[0.008, 0.147]$ | |
| | Relationship quality | 4 | 8 | 0.108 | $[-0.133, 0.336]$ | |
| Linkage direction | Bipolar | 13 | 69 | 0.073 | $[-0.019, 0.164]$ | 0.881 |
| | In-phase | 3 | 23 | 0.029 | $[-0.243, 0.297]$ | |
| | Pooled together | 5 | 50 | 0.050 | $[-0.032, 0.132]$ | |
| Lag considered | No | 9 | 53 | 0.082 | $[-0.025, 0.187]$ | 0.473 |
| | Yes | 8 | 89 | 0.048 | $[-0.031, 0.128]$ | |
| Temporal dependence adjusted | No | 14 | 133 | 0.077 | $[-0.011, 0.163]$ | 0.081 |
| | Yes | 1† | 9 | $-0.027$ | $[-0.154, 0.100]$ | |
| Linkage method family | Coherence based | 2 | 7 | $-0.112$ | $[-0.452, 0.256]$ | – |
| | Correlation based | 12 | 108 | 0.105 | $[-0.022, 0.227]$ | |
| | Local concordance | 1† | 1 | 0.355 | $[0.032, 0.611]$ | |
| | Model based coupling | 2 | 13 | 0.052 | $[-0.798, 0.833]$ | |
| | Pattern similarity | 2 | 13 | $-0.255$ | $[-0.547, 0.092]$ | |

$n$: distinct studies; $k$: effect sizes. Estimates and 95% CIs are back-transformed from Fisher's $z$ using single-moderator multilevel models with study-clustered CR2 inference. $p_{\text{mod}}$ tests differences across categories, not whether individual means differ from zero. Studies may contribute to multiple categories. †: one contributing study; interpret the estimate and interval cautiously. –: unavailable or nonfinite model output.

Table 4: Moderator analyses for unadjusted interaction effects.

| Context | Present | ANS | $n$ | $k$ | Estimated $r$ | 95% CI | $p_{\text{int}}$ |
|---|---|---|---|---|---|---|---|
| Stressful | No | PNS | 4 | 10 | $-0.158$ | $[-0.460, 0.177]$ | 0.282 |
| | Yes | PNS | 5 | 21 | $0.010$ | $[-0.117, 0.137]$ | |
| | No | SNS | 11 | 25 | $0.141$ | $[-0.063, 0.334]$ | |
| | Yes | SNS | 3 | 11 | $0.047$ | $[-0.656, 0.707]$ | |
| Helping | No | PNS | 8 | 27 | $-0.075$ | $[-0.174, 0.026]$ | 0.716 |
| | Yes | PNS | 1† | 4 | $0.173$ | $[0.032, 0.307]$ | |
| | No | SNS | 8 | 20 | $0.072$ | $[-0.135, 0.274]$ | |
| | Yes | SNS | 6 | 16 | $0.253$ | $[-0.129, 0.570]$ | |
| Goal-oriented | No | PNS | 5 | 25 | $-0.030$ | $[-0.145, 0.086]$ | 0.240 |
| | Yes | PNS | 3 | 6 | $-0.161$ | $[-0.537, 0.269]$ | |
| | No | SNS | 9 | 19 | $0.290$ | $[0.008, 0.529]$ | |
| | Yes | SNS | 5 | 17 | $-0.025$ | $[-0.254, 0.205]$ | |

$n$: distinct studies; $k$: effect sizes. Present indicates whether the context applies. Each context uses a separate ANS-by-context model. Estimates and 95% CIs are back-transformed from Fisher's $z$ with study-clustered CR2 inference. $p_{\text{int}}$ tests the interaction between ANS branch and context. Studies may contribute to multiple cells. †: one contributing study; interpret the estimate and interval cautiously. –: unavailable or nonfinite model output.